Single-move routing remains NP-complete on trees with candidate edge multiplicity eight

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Editorial note. Complete original mathematical paper102–2998, every reduction/algorithm auxiliary proof and all eight native proof figures retained unchanged. Original supplied main.bbl rendered verbatim as bibliography, since archive supplies bbl rather than bib; no reference reconstruction.

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Editorial note. Actual standalone wrapper interface mismatch: native LIPIcs author expects five arguments. Original author/affiliation/running/CCS/keyword entries36–53 are supplied unchanged, then only the wrapper one-argument author call is ignored, retaining those original native entries. Native class and all mathematics unchanged.

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Allowed standard prerequisites (editorial).

- Single-move labeled token routing, compatible matchings, precedence digraphs and parameterized reductions. Every native source/target label-set, route-emptiness and move-order convention retained. Source-defined solution edge multiplicity, candidate edge multiplicity and candidate vertex multiplicity remain distinct. All original reduction/DP correctness proofs and original figures included; external hardness inputs retain their original citations.. Full original mathematical paper and native main.bbl accompany each selected target; related variants are bundled and unselected simpler algorithms remain context.

1 Introduction

Neutral-atom quantum simulators and computers use optical traps to hold individual atoms. The goal is to place atoms in the *target traps*, a chosen subset of the optical traps; the interactions that define the quantum system occur only between nearby atoms, so the resulting arrangement of atoms, the *target configuration*, determines the structure of the system being simulated or used for computation [4–7]. Loading atoms directly into the target traps may not succeed, as the loading process does not guarantee that every target trap receives an atom. To increase the chances of obtaining the target configuration, one may instead load atoms into a larger set of traps and then rearrange the atoms that were successfully loaded. Some atoms may already occupy target traps and need not move, while other atoms are moved from occupied traps to empty target traps, each along a path of traps that are empty when the atom is moved. Computing the rearrangement, a task known as Atom Reconfiguration, requires choosing which atoms should move to which target traps, choosing routes for the moving atoms, and choosing a movement order so that no atom blocks a route when that route is used [7–10]. Each additional move takes time and creates another opportunity for errors or atom loss, which motivates keeping the number of moves of each atom small.

Atom Reconfiguration admits a graph-theoretic formulation. Vertices represent optical traps, source vertices are initially occupied by tokens representing atoms, and target vertices correspond to positions in the target configuration. A solution assigns sources to targets, chooses a route for each assigned pair, and orders the corresponding moves. Every move takes a token from its current vertex to an unoccupied vertex along a path no other vertex of which is occupied, the target included [1]. In general, a token may be moved several times. We study the restriction in which each token moves at most once.

Atoms that are not interchangeable. In Atom Reconfiguration, atoms are interchangeable: any loaded atom may be moved into any empty target trap. This is no longer the case as soon as atoms of two *species*, that is, of two chemical elements or of two isotopes of one element, are loaded together, as in dual-species settings, also called mixed-species settings [11–14]. Mixtures of two species are loaded deliberately, for instance to form a rubidium-caesium molecule from one atom of each species at every designated site [15, 16]. In these settings, a target trap may be designated for one species, so an atom of the wrong species is not an acceptable occupant [13, 14]. Thus, exchanging two atoms may turn a valid target configuration into an invalid one.

Restricting, for each target trap, which atoms may occupy it gives rise to the problem studied in this paper, Single-Move Labeled Token Routing, in which each source and each target is assigned a set of labels, and a source may be matched to a target if and only if their label sets intersect; in this case the source and the target are said to be *compatible*. Assigning the same label to every source and every target yields Atom Reconfiguration restricted to at most one move per atom, which we call the *all-matching case*; whether a rearrangement exists can then be decided in polynomial time [1, 8, 17]. At the opposite extreme, which we call the *unique-matching case*, every source is compatible with exactly one target and every target with exactly one source. Between these two extremes, the set of compatible source-target pairs can be arbitrary.

Our goal is to understand how the labels assigned to sources and targets, together with the structure of the underlying graph, affect the complexity of Single-Move Labeled Token Routing. We study several graph classes and parameterizations, proving both negative and positive results that focus on one question: how much information must an algorithm carry across a cut of the graph? Bounding the treewidth, or even the pathwidth,

of the graph does not make the problem tractable, since the problem is hard even on graphs of pathwidth three [2]. As we prove in this paper, bounding how many routes of a solution may share an edge does not either, since the problem is hard on grids even when the routes of a solution can be chosen pairwise edge-disjoint. Our main algorithmic result concerns trees, where every edge is a cut: bounding, for every edge, the number of compatible source-target pairs whose path uses that edge still does not make the problem tractable, but adding a bound on the maximum degree does, as does bounding, for every vertex, the number of compatible pairs whose paths contain that vertex.

The paper is organized as follows. Section 2 discusses related work and Section 3 presents an overview of our results. Section 4 introduces the necessary notation. Section 5 proves the hardness results, Section 6 the algorithmic ones, and Section 7 concludes with open problems.

2 Related work

Atom Reconfiguration, the problem motivating our work, is the setting in which every atom may move to every empty target trap, and in which one typically seeks to fill every target trap while minimizing some measure of the rearrangement. Atom reconfiguration has been carried out experimentally for one-, two-, and three-dimensional target configurations [4–7, 18]. On the algorithmic side, the assignment of atoms to target traps was first formulated as a minimum-cost matching problem [19]. Later algorithms treat the routes and the movement order as parts of the problem in their own right: after the targets and the routes are chosen, an order in which to execute the moves must still be found [8–10]. All of these algorithms assume interchangeable atoms and allow each atom to move more than once. The measures being minimized increase with the number of moves. Whenever the target traps can be filled at all, they can be filled by moving every atom at most once [1], a fact we reprove in our terminology. The all-matching case of our problem is exactly Atom Reconfiguration with as many atoms as target traps, restricted to at most one move per atom. Deciding whether every target trap can then be filled takes polynomial time (Theorem 1). Reconfiguration algorithms for two species of atoms have also been proposed [20,21]. The parameterized complexity of atom reconfiguration is known for interchangeable atoms and for the case in which each atom has its own target: most tractability results for the former become hardness results for the latter [17]. In this paper, we study what lies between these two cases: each atom may be compatible with an arbitrary set of targets.

Other models of token movement differ from ours in what a single move is and in which targets a token may be assigned to. We use the word *token* as a generic term for a moving object; the works below call their objects pebbles, agents, robots, or coins. Our move, along a path no vertex of which other than its first is occupied, was studied by Călinescu, Dumitrescu, and Pach [1] for interchangeable tokens, with the objective of minimizing the number of moves. They prove that at most k moves suffice, where k is the number of tokens, and that such a sequence can be chosen with no token moving twice. They also prove that minimizing the number of moves is NP-hard even on grids. A sequence of moves in which no token moves twice is a solution of our all-matching case, so their setting and ours differ only in the objective: they minimize the number of moves, whereas we decide feasibility. In Pebble Motion, a move takes a token along a single edge instead, and feasibility is decidable in linear time [22–25]. In other models, several tokens move along edges at the same time [26–31], or two tokens are swapped along the edge joining them [32–35]. The assignment of tokens to targets takes one of three forms: it is fixed, each token having its

own target; it is given by *colors*, each token and each target carrying one color and a token being assignable to every target of its color [3, 25, 36, 37]; or each token carries a set of targets to which it may be assigned [37,38]. Label sets express exactly the third form, which contains the other two. To our knowledge, arbitrary compatibility relations have not been studied before for moves along paths of unoccupied vertices.

In our problem, every token moves at most once; in the robotics literature, a sequence of moves with this property is called *monotone* [39]. Three results on monotone sequences of moves relate directly to ours. First, when both the matching and the routes are given as part of the input, feasibility is decided in polynomial time by an acyclicity test on a precedence digraph [38,40]; we state this as Observation 10 and use it in Section 6. Second, when the matching is given but the routes are not, the problem is hard. Geft, Halperin, and Nakar [2] study Monotone Sliding Reconfiguration, whose discrete form is exactly our unique-matching case, in which every source is compatible with exactly one target and every target with exactly one source. They prove it NP-complete on $3 \times w$ grid graphs, and NP-hard even when every token but one has a specified route. Section 5.2 shows that the hardness persists on instances that, when solvable, admit a solution with pairwise edge-disjoint routes. Third, in work independent of and concurrent with ours, Geft [3] proves that it is NP-hard to decide whether a Pebble Motion instance with two colors on a tree of maximum degree three can be solved with as many moves as the sum of the distances from the sources to the targets. In our terms, a color is a label set of size one, so such an instance uses two distinct labels. Geft further proves that a solution attaining this bound can be reordered so that each token traverses the path from its source to its target in consecutive moves, during which no other token occupies that path; such a traversal is exactly one move of our model. Hence, Single-Move Labeled Token Routing is NP-complete on trees of maximum degree three even when only two distinct labels are used.

3 Our contributions

We study *feasibility* throughout: deciding whether the target configuration can be reached at all. The statements below use the notation of Section 4: an instance $I = (G, S, T, L)$ consists of a graph G with n vertices and m edges, a set S of sources, a set T of targets, and a labeling L, which assigns a set of labels to each source and to each target, and $|I|$ denotes the length of its encoding.

We begin with the previously studied all-matching case, in which every source is compatible with every target. An all-matching instance is solvable if and only if every component of the graph contains as many sources as targets [1]. We provide a direct proof in our terminology; the proof simultaneously constructs a matching, a route for each matched pair, and an order of the moves by repeatedly removing leaves from a spanning tree of the graph.

▶ **Theorem 1.** *Single-Move Labeled Token Routing on all-matching instances can be solved, and a solution constructed, in time $\mathcal{O}(n^2 + m)$, and every connected all-matching instance is a yes-instance.*

At the other extreme, in the unique-matching case, every source is compatible with exactly one target and every target with exactly one source. This case is NP-complete on $3 \times w$ grid graphs [2], which have pathwidth at most three; in the instances produced by their reduction, every solution has an edge that lies on at least one third of the routes (Section 5.2). Our first two results give a construction whose instances, whenever they are solvable, admit a solution with pairwise edge-disjoint routes; the *solution edge multiplicity* of a solution is

the largest number of its routes that use a common edge, so such a solution has solution edge multiplicity one. Neither bounding the pathwidth nor bounding the solution edge multiplicity of the sought solution therefore makes the problem tractable. Our construction also lowers the maximum degree from four to three, at the price of allowing some edges to be shared by two routes of a solution.

▶ **Theorem 2.** *Single-Move Labeled Token Routing on unique-matching instances is NP-complete on planar graphs of maximum degree four, even on instances that, when solvable, admit a solution with pairwise edge-disjoint routes, and on planar graphs of maximum degree three, even on instances that, when solvable, admit a solution with solution edge multiplicity at most two.*

Grids and grid graphs are the graph classes closest to atom reconfiguration: one- and two-dimensional target configurations place target traps at regularly spaced positions, and moves between neighboring traps induce a grid-like graph structure. The same reduction transfers to them with minor modifications.

▶ **Theorem 3.** *Single-Move Labeled Token Routing on unique-matching instances is NP-complete on grids and on grid graphs, even on instances that, when solvable, admit a solution with pairwise edge-disjoint routes.*

On trees, unique-matching instances are solvable in polynomial time: the route of every pair is forced, and whether those routes can be ordered is decided by the acyclicity test described in Section 2. Trees are therefore where the label constraints are worth examining closely. By the independent and concurrent result discussed in Section 2, the problem is NP-complete on trees of maximum degree three even with two distinct labels [3]. We study the parameterized complexity of the problem on trees, taking as parameter an upper bound μ, given as part of the input, on the solution edge multiplicity of the sought solution. With this parameter, the problem is not fixed-parameter tractable unless $\mathsf{FPT} = \mathsf{W}[1]$, even when a movement order is given as part of the input.

▶ **Theorem 4.** *On trees of maximum degree three, Single-Move Labeled Token Routing is NP-complete, and, for a nonnegative integer μ given as part of the input, deciding whether there is a solution with solution edge multiplicity at most μ is W[1]-hard when parameterized by μ; both results hold even when a movement order is given as part of the input.*

The solution edge multiplicity counts only the routes of a solution. The *candidate edge multiplicity* is determined by the instance alone: for each edge, one counts the compatible source-target pairs whose unique paths in the tree use that edge, and the candidate edge multiplicity of the instance is the maximum of these counts over all edges. Our last hardness result shows that bounding the candidate edge multiplicity does not make the problem tractable either: the problem remains NP-complete when the candidate edge multiplicity is bounded by a constant, so unless $\mathsf{P} = \mathsf{NP}$, it admits no algorithm whose running time is polynomial in the input size for each fixed value of this parameter.

▶ **Theorem 5.** *Single-Move Labeled Token Routing is NP-complete on trees with candidate edge multiplicity at most eight, even when every source is compatible with at most three targets and every target with at most three sources.*

Next, we turn to algorithmic results. The first concerns paths, where the linear order of the vertices forces the matching: the ith source must be matched to the ith target, and it remains to test the compatibility of these pairs.

▶ **Theorem 6.** *Single-Move Labeled Token Routing on a path can be solved in time* $\mathcal{O}(|I|)$.

Our remaining results are parameterized. Since the hardness results above involve choosing a matching together with a movement order, it is natural to parameterize by the number k of tokens, which bounds the search space for both choices: there are at most $k!$ matchings and at most $k!$ movement orders per matching. For each guessed matching and movement order, the remaining question of whether suitable routes exist can be answered efficiently.

▶ **Theorem 7.** *Single-Move Labeled Token Routing is fixed-parameter tractable when parameterized by the number k of tokens.*

Finally, we return to trees for our main algorithmic results. Theorem 4 shows that bounding the solution edge multiplicity of the sought solution, even together with the maximum degree, does not make the problem tractable, and Theorem 5 shows the same for the candidate edge multiplicity alone. Bounding the candidate edge multiplicity together with the maximum degree, however, does. Each edge of a tree separates the tree into two sides, and the pairs counted for that edge are exactly the compatible pairs with a source on one side and a target on the other. When the maximum degree is also bounded, only a bounded number of compatible pairs cross the edges incident with any one vertex, so the problem can be solved by dynamic programming over the tree.

▶ **Theorem 8.** *Single-Move Labeled Token Routing on a tree instance $I = (G, S, T, L)$ can be solved in time $2^{\mathcal{O}(\Delta \cdot \mathrm{cm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$, where Δ is the maximum degree of G and $\mathrm{cm}(I)$ is the candidate edge multiplicity of I. Consequently, the problem is fixed-parameter tractable when parameterized by $\Delta + \mathrm{cm}(I)$.*

Neither parameter of Theorem 8 can be omitted: by Theorem 5 the problem is NP-complete on trees whose candidate edge multiplicity is constant, so parameterizing by the candidate edge multiplicity alone does not yield fixed-parameter tractability unless P = NP, and by Theorem 4 the problem is NP-complete on trees of maximum degree three, so the same holds for the maximum degree alone. Conversely, by Theorem 8, no NP-hardness reduction to instances on trees can produce only instances of constant maximum degree and constant candidate edge multiplicity, unless P = NP.

The construction of Theorem 5 has a single vertex of unbounded degree, and that vertex lies on the paths of unboundedly many compatible pairs. Counting the compatible pairs at vertices instead of edges gives the *candidate vertex multiplicity*, the maximum over all vertices of the number of compatible source-target pairs whose unique paths contain that vertex, and the same dynamic programming algorithm shows that the problem is fixed-parameter tractable when parameterized by this quantity alone.

▶ **Theorem 9.** *Single-Move Labeled Token Routing on a tree instance $I = (G, S, T, L)$ can be solved in time $2^{\mathcal{O}(\mathrm{cvm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$, where $\mathrm{cvm}(I)$ is the candidate vertex multiplicity of I. Consequently, the problem is fixed-parameter tractable when parameterized by $\mathrm{cvm}(I)$.*

4 Preliminaries

4.1 Graph terminology

We use standard graph-theoretic terminology; see, e.g., Diestel [41]. All graphs are finite, simple, and undirected unless stated otherwise. For a graph G we write $V(G)$ and $E(G)$ for its vertex set and edge set, and we set $n = |V(G)|$ and $m = |E(G)|$; for a path P we write

$V(P)$ and $E(P)$. For a positive integer i we write $[i] = \{1, \ldots, i\}$, and for an instance I of a decision problem we write $|I|$ for the length of its encoding.

A *grid* is a graph whose vertex set is the Cartesian product of two finite nonempty sets of consecutive integers, two vertices being adjacent if and only if they are at Euclidean distance one. Its *width* w and *height* h are the sizes of the first and second of these sets, and we call it an $h \times w$ *grid*. A *grid graph* is a subgraph of a grid, and an $h \times w$ *grid graph* is a subgraph of an $h \times w$ grid. The *treewidth* of a graph measures how far the graph is from being a tree, and its *pathwidth* how far it is from being a path; we refer the reader to Diestel [41] for the formal definitions.

Let G be a planar graph. A *plane drawing* of G consists of an injective map assigning to each vertex of G a point of $\mathbb{R}^2$ and, to each edge $uv \in E(G)$, a simple curve from the point representing u to the point representing v, such that the interior of each edge curve is disjoint from the points representing vertices and from the interiors of all other edge curves. An *orthogonal planar representation* of G is a plane drawing in which every edge is represented by a simple polygonal path all of whose segments are horizontal or vertical. An orthogonal planar representation is an *orthogonal grid drawing* if the points representing vertices and the endpoints of all segments representing edges have integer coordinates. Every planar graph of maximum degree at most four admits an orthogonal grid drawing of polynomial size that can be computed in polynomial time [42, 43].

4.2 Instances, moves, and solutions

Let G be a graph and let $S, T \subseteq V(G)$ with $|S| = |T|$. The vertices of S are *source vertices*, the vertices of T are *target vertices*, and the vertices of $V(G) \setminus (S \cup T)$ are *ordinary vertices*. At any point in time, a vertex is *occupied* if it contains a token and *unoccupied* otherwise. Initially each source holds one token, so precisely the vertices of S are occupied. In figures, source vertices are drawn as filled discs, target vertices as hollow squares, ordinary vertices as hollow discs, and vertices that are both sources and targets as filled discs enclosed in squares.

Let Λ be a finite set of *labels*. A *labeling* is a pair $L = (L_S, L_T)$ of functions $L_S \colon S \to 2^\Lambda$ and $L_T \colon T \to 2^\Lambda$; the sets $L_S(s)$ and $L_T(t)$ are the *label sets* of the source s and of the target t. A source $s \in S$ and a target $t \in T$ are *L-compatible* if $L_S(s) \cap L_T(t) \neq \emptyset$; when L is clear from context we simply say *compatible*. The set of compatible pairs $(s, t) \in S \times T$ is the *compatibility relation* of L. A label $\lambda \in \Lambda$ is *private* if it belongs to the label set of exactly one source and of exactly one target. An *L-compatible matching* is a set $M \subseteq S \times T$ in which every source of S and every target of T appears in exactly one pair, and every pair of which is L-compatible.

For a pair $(s, t) \in S \times T$, a *route* from s to t (an *s-t route*) is a simple s-t path in G, possibly *trivial*, that is, consisting of the single vertex $s = t$; the word *route* designates the path along which a token is moved in a solution. A *route family* for a compatible matching M is a family $\Pi = \{P_{s,t} : (s, t) \in M\}$ in which each $P_{s,t}$ is an s-t route; for a pair $q = (s, t)$ of M we also write P_q for $P_{s,t}$, $\mathrm{s}(q) = s$, and $\mathrm{t}(q) = t$. A *movement order* for M is an ordering of the pairs of M.

At a given point in time, a route $P_{s,t}$ is *clear* if no vertex of $V(P_{s,t}) \setminus \{s\}$ is occupied at that time; in particular, the route of a pair with $s = t$ is always clear. A *move* takes the token occupying s to t along a clear s-t route; after the move, t is occupied and, if $s \neq t$, s is unoccupied. In this paper every token is moved at most once, whence the name of the problem. Let π be a movement order for M and let Π be a route family for M. The move of a pair $q = (s, t)$ of M is *feasible* if P_q is clear once all pairs preceding q in π have been

moved, at which point the occupied vertices other than s are precisely the targets of the pairs before q and the sources of the pairs after q. The order π is *feasible for* Π if the move of every pair of M is feasible. A *solution* consists of a compatible matching M, a route family Π for M, and a movement order feasible for Π.

> Single-Move Labeled Token Routing
>
> **Input:** A graph G, sets $S, T \subseteq V(G)$ with $|S| = |T|$, one token on each vertex of S, a finite set Λ of labels, and a labeling $L = (L_S, L_T)$ with $L_S \colon S \to 2^\Lambda$ and $L_T \colon T \to 2^\Lambda$.
>
> **Question:** Do there exist a compatible matching $M \subseteq S \times T$, a route family $\Pi = \{P_{s,t} : (s,t) \in M\}$ for M, and a movement order feasible for Π?

We write an instance as $I = (G, S, T, L)$, leaving the label set Λ implicit, and call it a *tree instance* if G is a tree.

The problem belongs to NP: a compatible matching, a route family for it, and a movement order form a certificate of size polynomial in $|I|$, and simulating the moves in that order, while maintaining the set of occupied vertices, checks in polynomial time that every move is feasible. For each NP-completeness result below we therefore prove only hardness.

4.3 Special cases and precedence digraphs

An instance is *all-matching* if every source is compatible with every target, and *unique-matching* if each source is compatible with exactly one target and each target with exactly one source.

Let M be a compatible matching and let Π be a route family for M. The *precedence digraph* $D(M, \Pi)$ has vertex set M. For distinct pairs $p = (s_p, t_p)$ and $q = (s_q, t_q)$ of M, an arc (p, q) means that the move of p must precede the move of q, and is added in two cases. If $s_p \in V(P_q)$, then the token at s_p must leave before the route P_q is used. If $t_q \in V(P_p)$, then the token moving to t_q may arrive only after the route P_p has been used.

The characterization below is the acyclicity test referenced in Section 2; we prove it in our notation because the dynamic programming algorithm of Section 6 relies on it.

▶ **Observation 10.** *Let M be a compatible matching and let Π be a route family for M. There exists a feasible movement order for Π if and only if $D(M, \Pi)$ is acyclic.*

Proof. Suppose that π is a feasible movement order for Π, and let (p, q) be an arc of $D(M, \Pi)$. If $s_p \in V(P_q)$, then s_p is a vertex of P_q other than s_q, occupied until p is moved, so p precedes q. If $t_q \in V(P_p)$, then either $t_q = s_p$, and the previous case applies, or t_q is a vertex of P_p other than s_p, occupied once q is moved, so again p precedes q. Hence, every arc goes from an earlier to a later pair of π, and $D(M, \Pi)$ has no directed cycle.

Conversely, suppose that $D(M, \Pi)$ is acyclic and let $\pi = (q_1, \ldots, q_k)$ be an ordering of the pairs of M in which every arc of $D(M, \Pi)$ goes from an earlier pair to a later pair; such a topological ordering exists because $D(M, \Pi)$ is acyclic. Consider a pair q_i of this ordering and write s_i for its source. If the route of q_i contains the source of some pair q_j with $j \neq i$ other than at s_i, then (q_j, q_i) is an arc of the precedence digraph, so $j < i$ and that source has already been vacated. If the route of q_i contains the target of some pair q_j with $j \neq i$ other than at s_i, then (q_i, q_j) is an arc, so $i < j$ and that target is not yet occupied. Hence, when the token at s_i is moved, every vertex of the route of q_i other than s_i is unoccupied: the route is clear. Since this holds for every i, the movement order π is feasible. ◀

4.4 Parameterized complexity and multiplicity

We use standard parameterized complexity terminology [44]. A parameterized problem is *fixed-parameter tractable* if there is a computable function f such that every instance I with parameter κ can be solved in time $f(\kappa) \cdot |I|^{\mathcal{O}(1)}$; the class of all fixed-parameter tractable problems is denoted FPT. A *parameterized reduction* from a parameterized problem Q to a parameterized problem Q' is an algorithm that maps every instance (I, κ) of Q to an instance (I', κ') of Q' that is a yes-instance if and only if (I, κ) is, and that does so in time $f(\kappa) \cdot |I|^{\mathcal{O}(1)}$ and with $\kappa' \leq f(\kappa)$, for some computable function f. A parameterized problem is *W*[1]*-hard* if every problem of the class W[1] admits a parameterized reduction to it; unless FPT = W[1], no W[1]-hard problem is fixed-parameter tractable [44].

The first of our three multiplicity measures is defined for route families in arbitrary graphs. For a route family Π, the *solution edge multiplicity* of Π is $\mathrm{sm}(\Pi) = \max_{e \in E(G)} |\{P \in \Pi : e \in E(P)\}|$. The solution edge multiplicity of a solution is that of its route family. In Theorem 4 we parameterize by an upper bound on this quantity: the input contains, in addition to an instance I, a nonnegative integer μ, the question is whether I admits a solution with solution edge multiplicity at most μ, and the parameter is μ.

The second measure is defined only for tree instances $I = (G, S, T, L)$, in which every pair $(s, t) \in S \times T$ is joined by a unique path $P_{s,t}$. For an edge $e \in E(G)$, the *candidate edge multiplicity* of e is $\mathrm{cm}_I(e) = |\{(s, t) \in S \times T : s \text{ and } t \text{ are compatible and } e \in E(P_{s,t})\}|$, and the *candidate edge multiplicity* of I is $\mathrm{cm}(I) = \max_{e \in E(G)} \mathrm{cm}_I(e)$. The third measure counts the same pairs at vertices instead of edges. For a vertex $v \in V(G)$, the *candidate vertex multiplicity* of v is $\mathrm{cvm}_I(v) = |\{(s, t) \in S \times T : s \text{ and } t \text{ are compatible and } v \in V(P_{s,t})\}|$, and the *candidate vertex multiplicity* of I is $\mathrm{cvm}(I) = \max_{v \in V(G)} \mathrm{cvm}_I(v)$. The name *multiplicity* is borrowed from the literature on coordinating the movement of robots along paths given as part of the input, where the *vertex multiplicity* of an instance is the maximum number of these paths through a single vertex [45].

For every route family Π of a solution of a tree instance I we have $\mathrm{sm}(\Pi) \leq \mathrm{cm}(I)$, since every route of a solution is the unique path of some compatible pair. The ratio between the two can nevertheless be linear in the number k of tokens. Consider an all-matching instance on a star whose leaves are k sources and k targets, with $S \cap T = \emptyset$. Every source is compatible with every target, and the path from a source leaf to any target leaf uses the edge incident with that source leaf. That edge therefore lies on the paths of k compatible pairs, and $\mathrm{cm}(I) = k$. In every solution each source is matched with exactly one target, so every edge lies on exactly one route of the solution, and every solution has solution edge multiplicity one.

4.5 The all-matching case

We close the preliminaries with the all-matching case, the starting point for our study of compatibility constraints. When the matching is unrestricted, the matching, the routes, and the movement order can be chosen simultaneously, by a procedure that repeatedly deletes a leaf of a spanning tree. The two nontrivial cases are a leaf that is a target, which is filled from a closest remaining source and whose move is placed *before* all moves produced later, and a leaf that is a source, which is emptied into a closest remaining target and whose move is placed *after* all moves produced later; in each case the minimality of the choice is what makes the route clear at the time it is used. The result below is due to Călinescu, Dumitrescu, and Pach [1], and a stronger form, in which the matching is additionally required to minimize the total distance traveled by the tokens, is also known [8].

▶ **Theorem 1.** *Single-Move Labeled Token Routing on all-matching instances can be solved, and a solution constructed, in time $\mathcal{O}(n^2 + m)$, and every connected all-matching instance is a yes-instance.*

Proof. Let I be an all-matching instance on a graph G with source set S and target set T, where $|S| = |T|$.

If G is disconnected, then every route lies in a single connected component, so every component must contain the same number of sources as targets; if this fails, the instance is a no-instance, and otherwise it suffices to solve the instance independently in each component and concatenate the resulting movement orders. We may therefore assume that G is connected.

Compute an arbitrary spanning tree F of G. Since every route in F is also a route in G, it is enough to construct a solution whose routes use only edges of F. We describe a recursive procedure on the current tree F together with the current source set S and target set T, maintaining the invariant that F is connected and $|S| = |T|$. Let v be a leaf of F.

1. **Case $v \notin S \cup T$.** The leaf v is irrelevant, as no route between two remaining vertices uses it. Delete v from F and continue.
2. **Case $v \in S \cap T$.** Match v to itself and use the trivial route. Since v is a leaf, it is not an internal vertex of any route between the remaining vertices, and its move may be placed anywhere in the movement order, since its route contains no vertex other than v itself. Delete v from F, remove it from S and from T, and continue.
3. **Case $v \in T \setminus S$.** Choose a source $s \in S$ closest to v in F, match s to v, and let P be the unique s-v path in F. Place this move before the moves obtained recursively; at the time it is executed, the occupied vertices are precisely the current sources. By the choice of s, no internal vertex of P belongs to S, and $v \notin S$, so s is the only occupied vertex of P and P is clear. Delete v from F, remove s from S and v from T, and continue.
4. **Case $v \in S \setminus T$.** Choose a target $t \in T$ closest to v in F, match v to t, and let P be the unique v-t path in F. By the choice of t, no internal vertex of P belongs to T. Place this move after the moves obtained recursively, delete v from F, remove v from S and t from T, and continue. After the recursive moves have been executed, the occupied vertices of the remaining tree are precisely the remaining targets; in particular t is still unoccupied and every internal vertex of P is unoccupied, so P is clear at that time and the token at v can be moved to t.

Each reduction removes at least one vertex. The case analysis shows how to extend a feasible movement order of the reduced instance to one of the original instance: in the case $v \in T \setminus S$ one prepends the move into v, and in the case $v \in S \setminus T$ one appends the move out of v. By induction on $|V(F)|$, the original instance admits a feasible movement order.

A spanning tree of G can be computed in time $\mathcal{O}(n + m)$. The recursive procedure performs at most n leaf removals, and with a straightforward implementation each step finds the closest remaining source or target by a graph search in time $\mathcal{O}(n)$. Listing the routes takes total time $\mathcal{O}(n^2)$, since at most n moves are produced and each route has length at most $n - 1$. Hence, the whole solution is constructed in time $\mathcal{O}(n^2 + m)$. ◀

5 Hardness under structural and compatibility constraints

We now prove the hardness results stated in Section 3. Their purpose is to separate several possible sources of difficulty in Single-Move Labeled Token Routing. Section 5.2 shows that the problem is hard on planar graphs of bounded degree and on grids, even when

the compatible matching is unique and the instances, when solvable, admit a solution with pairwise edge-disjoint routes. Hardness therefore arises neither from the need to choose a matching nor from any need for two routes to share an edge. We then turn to trees, where the route of every compatible pair is forced. Section 5.3 shows that on trees of maximum degree three, choosing a matching is W[1]-hard when parameterized by an input bound on the solution edge multiplicity, even when a movement order is given as part of the input, and Section 5.4 shows that on trees of unbounded degree, the problem is NP-complete even when the candidate edge multiplicity is at most eight.

5.1 Reducing the degree of ordinary vertices

For ease of understanding, we first describe each construction using ordinary vertices of high degree and then reduce the degree by the operation defined next, which replaces such a vertex by a tree of ordinary vertices of maximum degree three. The resulting instance is a yes-instance if and only if the original one is.

Let u be an ordinary vertex of a graph G with $d \geq 4$ neighbors, and let F_u be a tree of maximum degree three with d leaves and at most $2d$ vertices. The *expansion* of u *by* F_u replaces u by F_u: it deletes u, adds a disjoint copy of F_u whose vertices are all ordinary, and joins the d leaves of the copy to the d neighbors of u, one leaf to each neighbor. The resulting graph is denoted $G - u + F_u$.

▶ **Lemma 11.** *Let (G, S, T, L) be an instance of* Single-Move Labeled Token Routing, *let $u \in V(G)$ be an ordinary vertex of degree at least four, and let $G' = G - u + F_u$ be obtained by expanding u by a tree F_u of maximum degree three with as many leaves as u has neighbors. Then (G, S, T, L) is a yes-instance if and only if (G', S, T, L) is a yes-instance. Moreover, G' is a tree if G is a tree, and, if G is planar, the leaves can be attached to the neighbors of u so that G' is planar.*

Proof. Every vertex of F_u is ordinary, and is therefore permanently unoccupied. Consider a solution of (G, S, T, L). Since u is ordinary, every route that uses u contains a subpath of the form v, u, w, where v and w are neighbors of u in G. Replace this subpath by the path from v to w that enters F_u through the leaf adjacent to v, follows the unique path in F_u to the leaf adjacent to w, and then proceeds to w. This transforms every route into a route of G'. Since the newly introduced vertices are permanently unoccupied, every route that was clear in G remains clear in G'. Hence, the same matching and movement order, together with the transformed routes, form a solution of (G', S, T, L).

Conversely, consider a solution of (G', S, T, L), with route family Π'. Contract F_u to a single vertex u. Every route in Π' then becomes a walk in G. Since the original route is simple, the resulting walk can repeat only u. For each such walk, retain the subwalk from its source to its first occurrence of u and the subwalk from its last occurrence of u to its target, identifying these two occurrences as a single occurrence of u. The resulting walk repeats no vertex and is therefore a simple route with the same endpoints. Its internal vertices, apart from the permanently unoccupied vertex u, form a subset of the internal vertices of the original route. Therefore, every route that was clear in G' yields a clear route in G. Hence, the same matching and movement order, together with the resulting routes, form a solution of (G, S, T, L).

If G is a tree, then G' is obtained from the forest $G - u$ by attaching each of its components to a distinct leaf of F_u, so G' is a tree.

For planarity, fix a plane drawing of G and a closed disc around u so small that it contains no vertex other than u, meets no edge that is not incident with u, and meets each

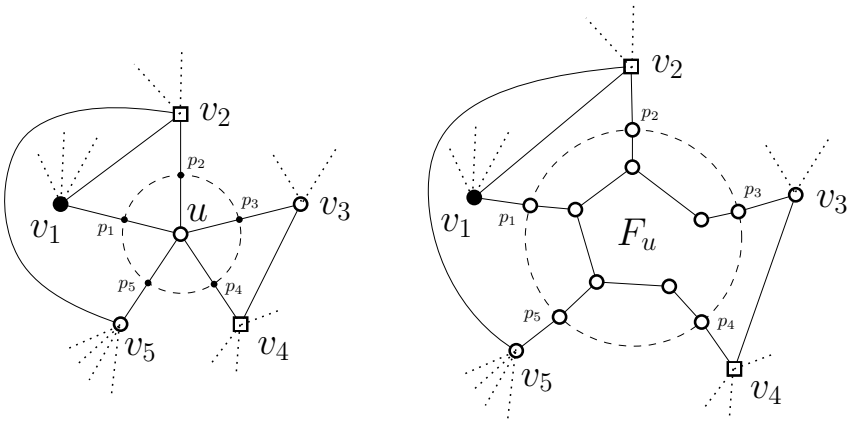


Figure 1 The expansion of an ordinary vertex u of degree at least four. On the left, u has five neighbors $v_1, \ldots, v_5$, and the edge uv_i crosses the boundary of the dashed disc around u at the point p_i. On the right, the part of the drawing inside the disc is replaced by a tree F_u of maximum degree three, whose leaves lie at the points $p_1, \ldots, p_5$, and the leaf at p_i is joined to v_i by the part of uv_i outside the disc; for readability, the disc is enlarged on the right and the rest of the drawing is stretched accordingly. None of the vertices of F_u is a source or a target. Dotted edge segments indicate that the graph continues beyond the neighbors of u: the rest of the graph, which the expansion leaves unchanged, is not drawn.

edge incident with u in a single curve from u to the boundary of the disc. Let $p_1, \ldots, p_d$ be the points at which these edges cross the boundary of the disc, in cyclic order, and let $v_1, \ldots, v_d$ be the corresponding neighbors of u. Draw F_u inside the disc without crossings, with its d leaves at the points $p_1, \ldots, p_d$, and join the leaf at p_i to v_i by the part of the edge uv_i outside the disc, for each $i \in [d]$. This yields a plane drawing of G' (Figure 1). ◀

5.2 Unique matching with edge-disjoint routes on planar graphs and grids

On unique-matching instances the matching is fixed, and Geft, Halperin, and Nakar [2] show that choosing the routes is hard on $3 \times w$ grid graphs, even when every token but one has a prescribed route. When the routes are not prescribed, every solution of their instances has an edge that lies on at least $k/3$ of the routes, where k is the number of tokens: all sources lie to the left of all targets, so every route crosses any vertical line between them, and only three edges cross such a line, so one of them lies on at least a third of the routes. We show that hardness does not require routes that share edges: every instance constructed in this subsection that is a yes-instance admits a solution whose routes are pairwise edge-disjoint, that is, a solution with solution edge multiplicity one.

This subsection proves Theorems 2 and 3, both by modifying a single reduction that we first present on general graphs. On general graphs the reduction yields a weaker statement than the known hardness on $3 \times w$ grid graphs, but its construction and correctness proof are the basis of both modifications, so we give them in full.

5.2.1 The reduction on general graphs

We reduce from Planar Monotone Rectilinear 3-SAT; we first define its instances. A clause of a CNF formula is *positive* if none of its literals are negated variables and *negative*

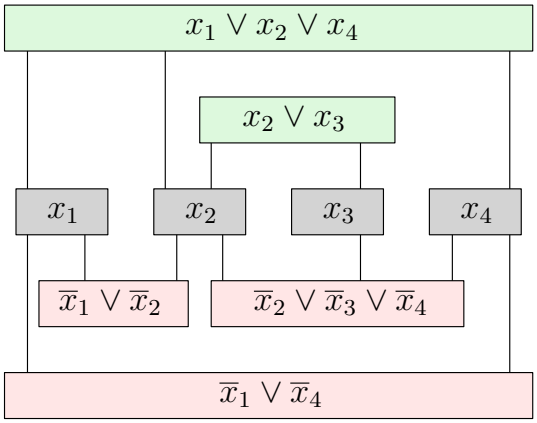


Figure 2 A rectilinear representation of the Planar Monotone Rectilinear 3-SAT instance $\varphi = (x_1 \vee x_2 \vee x_4) \wedge (x_2 \vee x_3) \wedge (\overline{x}_1 \vee \overline{x}_2) \wedge (\overline{x}_2 \vee \overline{x}_3 \vee \overline{x}_4) \wedge (\overline{x}_1 \vee \overline{x}_4)$. The positive clauses are drawn above the variable line, and the negative clauses below it. A clause box is joined by a vertical segment to each of its variables, so one clause may reach several variables.

if all of them are. A formula is *monotone* if each of its clauses is positive or negative. A *rectilinear representation* of a monotone formula φ consists of an axis-parallel rectangle for each variable and for each clause of φ, called the *box* of that variable or clause, and of a vertical segment for each clause C and each variable occurring in C, which joins the box of C to the box of that variable, such that the boxes are pairwise disjoint, the boxes of the variables all meet one horizontal line, called the *variable line*, the box of every positive clause lies above that line and the box of every negative clause below it, and every vertical segment meets the two boxes it joins only at its endpoints and meets no other box and no other vertical segment (Figure 2). We identify each variable and each clause with its box.

An instance of Planar Monotone Rectilinear 3-SAT is a monotone CNF formula whose clauses contain at most three literals, together with a rectilinear representation of the formula; the question is whether the formula is satisfiable. This problem is NP-complete [46].

Let $(\varphi, \mathcal{R})$ be such an instance, so that φ is a monotone CNF formula whose clauses contain at most three literals and $\mathcal{R}$ is a rectilinear representation of φ. Throughout the construction, x denotes a variable of φ and C a clause of φ. We assume that every clause contains two or three literals, that no two literals of a clause have the same variable, and that every variable occurs in some clause. All three assumptions can be made without loss of generality. A clause with a single literal can only be satisfied by setting its variable accordingly. Doing so, removing every clause that becomes satisfied, deleting the literal that becomes false from every remaining clause, and repeating this while some clause has a single literal yields, in polynomial time, a formula that has no such clause and is satisfiable if and only if φ is, unless a clause becomes empty, in which case φ is unsatisfiable; its rectilinear representation is obtained from $\mathcal{R}$ by deleting the boxes and segments of the removed clauses and literals. In a monotone formula a variable occurring twice in a clause occurs twice as the same literal, so removing the repeated literal leaves an equivalent formula with the same rectilinear representation. A variable that occurs in no clause can be deleted together with its box. An *occurrence* is a pair consisting of a clause and one of the literals of that clause, called the *literal of the occurrence*; the *variable of the occurrence* is the variable of its literal, and we say that the occurrence is an occurrence *of* that variable and *in* that clause. Occurrences in distinct clauses are distinct even when their literals are equal, so the number

of occurrences is the sum of the sizes of the clauses. We identify a clause C with the set of its occurrences and write $o \in C$. An occurrence o is *positive* if its literal is its variable and *negative* if its literal is the negation of its variable, and o is *true* or *false* under a truth assignment according to the value of its literal.

To represent φ as a graph, we create tokens and gadgets for each variable and each clause. A variable gadget contains two paths for the *assignment token*, one corresponding to setting the variable to `true` (the *true path* for the variable), and the other to setting the variable to `false` (the *false path* for the variable). In a clause gadget, there are at most three possible paths for the *clause token*, one corresponding to each occurrence in the clause.

Additional tokens tie the paths available to a clause token to the paths taken by the assignment tokens. For each occurrence, a *blocking token* has its source in the variable gadget, on the true path for a negative occurrence and on the false path for a positive occurrence, and its target on the path of the clause gadget associated with that occurrence. When the assignment token moves along the true path or the false path, every blocking token whose source lies on that path must move out of its way, into the clause gadget of its occurrence, where it blocks the path of that occurrence. For each occurrence, a *delay token* has its source in the clause gadget, on every path from the source of the clause token to its target, and its target in the variable gadget, on the same path as the source of the blocking token. The clause token therefore moves only after the assignment tokens that pass the targets of its delay tokens, by which time the blocking tokens on the paths of those assignment tokens have reached the clause gadget.

Construction. For a token z, we write $\mathrm{s}(z)$ and $\mathrm{t}(z)$ for its source and its target. Throughout this subsection the letters a, b, c, and d, with subscripts, denote tokens.

We make use of the following types of tokens, each named by the initial of its role: an *assignment token* a_x for each variable x, a *blocking token* b_o and a *delay token* d_o for each occurrence o, and a *clause token* c_C for each clause C. When the occurrences of one clause are numbered $o_1, o_2, \ldots$, we write b_i and d_i for the blocking token and the delay token of o_i.

Figure 3 shows the part of the construction arising from one clause with three occurrences and from the three variables occurring in it. We describe the variable gadgets, the clause gadgets, and the occurrence edges in turn.

Variable gadgets. For each variable x, we create the vertices $\mathrm{s}(a_x)$ and $\mathrm{t}(a_x)$, an ordinary vertex g_x, called the *junction* of the gadget, and, for each occurrence o of x, the vertices $\mathrm{s}(b_o)$ and $\mathrm{t}(d_o)$. The *true path* is a path from g_x to $\mathrm{s}(a_x)$ whose internal vertices are $\mathrm{s}(b_o)$ followed by $\mathrm{t}(d_o)$ for each negative occurrence o of x, the pairs of distinct occurrences appearing in an arbitrary order; the *false path* is a path from g_x to $\mathrm{s}(a_x)$ defined in the same way from the positive occurrences of x; the *variable tail* is the edge $g_x\mathrm{t}(a_x)$. Sources and targets thus alternate along the true path and along the false path.

Clause gadgets. For each clause C, we create the vertices $\mathrm{s}(c_C)$ and $\mathrm{t}(c_C)$, an ordinary vertex g_C, called the *junction* of the gadget, and, for each occurrence $o \in C$, the vertices $\mathrm{t}(b_o)$ and $\mathrm{s}(d_o)$. The *literal path* of o is the path $\mathrm{s}(c_C), \mathrm{t}(b_o), g_C$, and the *clause tail* is a path from g_C to $\mathrm{t}(c_C)$ whose internal vertices are the vertices $\mathrm{s}(d_o)$ with $o \in C$, in an arbitrary order.

Occurrence edges. For each occurrence o, we add the edge $\mathrm{s}(b_o)\mathrm{t}(b_o)$ and the edge $\mathrm{s}(d_o)\mathrm{t}(d_o)$, the *occurrence edges* of b_o and of d_o.

Labels. For each token, we assign a private label to its source and its target, so that the compatible matching of the constructed instance is unique.

In this reduction, the variable gadgets and clause gadgets are pairwise vertex-disjoint,

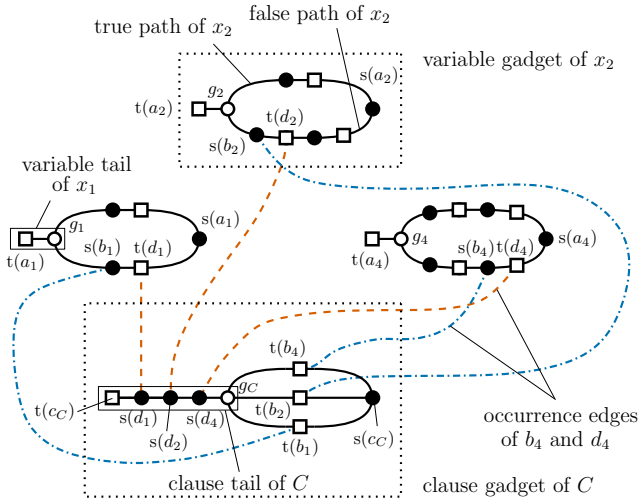


Figure 3 Part of the construction for the positive clause $C = x_1 \vee x_2 \vee x_4$ of φ (Figure 2); for $i \in \{1, 2, 4\}$, a_i and g_i denote the assignment token and the junction of the gadget of x_i, and b_i and d_i the blocking and delay tokens of the occurrence of x_i in C. In every variable gadget, the true path is drawn above and the false path below. The dotted rectangles single out the gadget of x_2 and the gadget of C. The blue dash-dotted curves and the orange dashed curves are the occurrence edges of the blocking tokens and of the delay tokens, respectively. The occurrence edges of other clauses are omitted, but the sources and targets that those clauses create on the true and false paths are shown.

and each occurrence edge meets the two gadgets it joins only at its endpoints. Let $J = (G, S, T, L)$ denote the constructed instance of Single-Move Labeled Token Routing. Each variable, clause, and occurrence contributes a constant number of vertices and edges, so J can be computed in polynomial time. We next present the properties of the instance J that will be used in the correctness proof. The first observation shows that the occurrence edge of a blocking or delay token is used by no other token; since these edges are the only edges joining two gadgets, the assignment tokens and the clause tokens are confined to their own gadgets, which leaves each of them the few routes described in the second observation.

▶ **Observation 12.** *Let o be an occurrence and let z be b_o or d_o. In every solution of J, no route of a token other than z contains both $\mathrm{s}(z)$ and $\mathrm{t}(z)$; in particular, no such route uses the occurrence edge of z.*

Proof. Fix an occurrence o and let z be b_o or d_o. At every point of a movement order, the vertex $\mathrm{s}(z)$ is occupied if z has not yet moved, and the vertex $\mathrm{t}(z)$ is occupied otherwise, so at least one of the two is occupied at every point. Neither vertex is the source or the target of a token other than z, so neither is the first vertex of the route of another token. A route of another token containing both therefore contains, at the time it is used, an occupied vertex other than its own first vertex, and is not clear. The occurrence edge of z joins these two vertices, so no route of another token uses it. ◀

▶ **Observation 13.** *In every solution of J, the route of each assignment token a_x consists of either the true path or the false path of the variable gadget for x, followed by its variable tail. Similarly, the route of each clause token c_C consists of one literal path of the clause gadget for C, followed by its clause tail.*

Proof. By Observation 12, neither an assignment token nor a clause token can use an occurrence edge. Since the gadgets are otherwise vertex-disjoint, the route of an assignment token remains inside its variable gadget, and the route of a clause token remains inside its clause gadget. The structure of the routes follows from the structure of the two gadgets. ◀

Confining the assignment tokens and the clause tokens to their gadgets also settles the solution edge multiplicity of J.

▶ **Observation 14.** *In every solution of J, the routes of the tokens are pairwise edge-disjoint. Consequently, every solution of J has solution edge multiplicity one.*

Proof. By Observation 13, the route of an assignment token uses only edges of its variable gadget, and the route of a clause token only edges of its clause gadget; the gadgets are pairwise vertex-disjoint, so these routes are pairwise edge-disjoint. By Observation 12, the occurrence edge of a blocking or delay token z is used by no token other than z. The token z cannot use any other edge, since every route from $\mathrm{s}(z)$ to $\mathrm{t}(z)$ other than its occurrence edge would have to leave the gadget containing $\mathrm{s}(z)$ through the occurrence edge of another token. Hence, every edge belongs to at most one route. ◀

The remaining property concerns the order in which the tokens move. Consider a solution of J with route family Π. By Observation 13, every assignment token a_x uses either the true path or the false path of its gadget, and we let τ_Π be the truth assignment that sets x to `true` if a_x uses its true path and to `false` if a_x uses its false path. For the reverse direction of the reduction, we will show that τ_Π satisfies φ; the observation below gives the precedence constraints from which this follows.

▶ **Observation 15.** *Let o be an occurrence in a clause C. In every solution of J with route family Π, if o is false under τ_Π, then $\mathrm{t}(b_o)$ is occupied when c_C moves.*

Proof. Let x be the variable of o. By construction, $\mathrm{s}(b_o)$ and $\mathrm{t}(d_o)$ lie on the false path of the variable gadget for x if o is positive and on its true path if o is negative, so both lie on the route of a_x precisely when o is false under τ_Π. Suppose that o is false. The source $\mathrm{s}(b_o)$ must be vacated before a_x moves, so b_o moves before a_x. Moving d_o before a_x would occupy $\mathrm{t}(d_o)$, an internal vertex of the route of a_x, so a_x moves before d_o. By Observation 13, the route of c_C contains the clause tail, and $\mathrm{s}(d_o)$ lies on this tail, so d_o moves before c_C. Therefore, b_o moves before c_C, and since each token moves at most once, $\mathrm{t}(b_o)$ is occupied when c_C moves. ◀

We now have all the ingredients needed to establish the correctness of the reduction.

▶ **Lemma 16.** *Single-Move Labeled Token Routing on unique-matching instances is NP-complete, even on instances in which the routes of every solution are pairwise edge-disjoint.*

Proof. We show that φ is satisfiable if and only if J is a yes-instance.

($\Rightarrow$) Suppose that φ has a satisfying assignment τ. First, for every occurrence o that is false under τ, move the blocking token b_o along its occurrence edge to $\mathrm{t}(b_o)$. This edge has no internal vertex, and $\mathrm{t}(b_o)$ is unoccupied, since no token has yet reached a target, so every such move is feasible.

Next, for every variable x, move a_x along its true path if $\tau(x) =$ `true` and along its false path if $\tau(x) =$ `false`, followed by its variable tail. The only sources on the chosen path are those of the blocking tokens of the occurrences of x that are false under τ, and all

of these sources were vacated in the first step. No source of another assignment token, of a clause token, or of a delay token lies on the chosen path, since the gadgets are pairwise vertex-disjoint and the sources of clause and delay tokens lie in clause gadgets. Moreover, no delay token has yet moved, so every target of a delay token on the chosen path is unoccupied, and so is $\mathrm{t}(a_x)$. Hence, the route of every assignment token is clear.

We next move every delay token along its occurrence edge, which has no internal vertex. Its target is unoccupied, since no other token is assigned to it, and since every assignment token has already reached its target, occupying it does not block any route that remains to be used.

Then, consider a clause C. Since τ satisfies C, there exists an occurrence $o \in C$ that is true under τ. Only the blocking tokens of false occurrences moved in the first step, so b_o has not moved, and $\mathrm{t}(b_o)$ is unoccupied. Every delay token has moved, so every source of a delay token on the clause tail of C has been vacated, and $\mathrm{t}(c_C)$ is unoccupied. Thus, c_C can move along the literal path of o and then along the clause tail to $\mathrm{t}(c_C)$.

Finally, move every remaining blocking token along its occurrence edge, which has no internal vertex; its target is unoccupied, since no other token is assigned to it. Every token has moved along a clear route, so the unique compatible matching of J, the routes along which the tokens moved, and this movement order form a solution of J.

($\Leftarrow$) Suppose that J is a yes-instance, and let Π be the route family of a solution. We show that τ_Π, the truth assignment that sets x to `true` if a_x uses its true path and to `false` if a_x uses its false path, satisfies φ. Consider a clause C and suppose, for a contradiction, that every occurrence in C is false under τ_Π. By Observation 15, the target of the blocking token of each occurrence in C is occupied when c_C moves. By Observation 13, the route of c_C uses a literal path of the clause gadget for C, and each literal path contains the target of the blocking token of its occurrence. Hence, whichever literal path the route of c_C uses, the route contains a vertex other than its source that is occupied when c_C moves, so no route available to c_C is clear at that time. This contradicts the feasibility of the movement order, so τ_Π satisfies C; hence τ_Π satisfies every clause of φ.

By Observation 14, the routes of every solution of J are pairwise edge-disjoint. Together with membership in NP, established in Section 4, this proves the lemma. ◀

5.2.2 A planar realization

The graph of J need not be planar. We now modify J, in polynomial time, into an instance $J_{\mathrm{pl}} = (G_{\mathrm{pl}}, S, T, L)$ whose graph is planar. The modification changes the clause gadgets and fixes the order of the vertices along the true and false paths of the variable gadgets; Figure 4 shows the resulting graph for the formula of Figure 2, drawn as described in the proof of Lemma 18 below.

Both the new clause gadgets and the order of the vertices along the paths of the variable gadgets are determined by the rectilinear representation $\mathcal{R}$ given with φ. The *segment* of an occurrence o is the vertical segment of $\mathcal{R}$ that joins the box of the clause of o to the box of the variable of o. Its endpoint on the box of the clause is the *clause port* of o, which lies on the side of that box that faces the variable line, and its endpoint on the box of the variable is the *variable port* of o, which lies on the upper side of that box if o is positive and on its lower side if o is negative. Distinct occurrences have distinct segments, since no two literals of a clause have the same variable, and the segments of distinct occurrences are disjoint. Hence, the clause ports of a clause are distinct points of one horizontal side of its box, and so are the variable ports of the positive occurrences of a variable, and those of its negative occurrences. We order the ports on a side from left to right, that is, by increasing

x-coordinate. We number the occurrences of each clause C as $o_1, \ldots, o_{|C|}$ in the order of their clause ports, writing b_i and d_i for the tokens of o_i as in Section 5.2.1, and we order the positive occurrences of each variable, and separately its negative occurrences, in the order of their variable ports.

The correctness of J rests on two properties (Observations 13 and 15): the route of an assignment token is its true path or its false path, followed by its variable tail, and the route of a clause token c_C contains $\mathrm{t}(b_o)$ and $\mathrm{s}(d_o)$ for some occurrence $o \in C$, which forces b_o to block the route of c_C when o is false. The modification preserves both properties. Its purpose is to allow a plane drawing of the resulting graph, which we now describe.

We call a vertex of a gadget an *anchor* if an occurrence edge ends at it. In the drawing, each gadget lies inside its box, and the two occurrence edges of each occurrence o run alongside the segment of o, that of b_o to its left and that of d_o to its right, crossing the side of each of the two boxes immediately to the left and to the right of the port of o on that side. The occurrence edges entering a box therefore arrive at each of its sides in the order of the ports on that side, the edge of b_o immediately followed by that of d_o. Inside the box, each of these edges is completed by a straight line segment to its anchor. These segments cross neither each other nor the gadget if the anchors of the edges arriving at a side lie on a line parallel to that side, in the order in which the edges arrive, and no point of the gadget lies strictly between this line and that side: the segments then join points of two parallel lines in the same order, and the gadget has no point between these lines.

In a variable gadget, the anchors on the upper side of the box are the vertices $\mathrm{s}(b_o)$ and $\mathrm{t}(d_o)$ of the positive occurrences o, all on the false path, and those on the lower side are the corresponding vertices of the negative occurrences, all on the true path. The variable gadget can therefore be drawn as required once these vertices appear along the false path in the order of the positive occurrences, and along the true path in the order of the negative ones; this is how the modification fixes the order of a variable gadget's vertices.

In the gadget of a clause C, the anchors must lie on the line in the order $\mathrm{t}(b_1)$, $\mathrm{s}(d_1)$, $\ldots$, $\mathrm{t}(b_{|C|})$, $\mathrm{s}(d_{|C|})$. The clause gadget of J cannot be drawn in this way, since every delay source lies on the clause tail and every blocking target on a literal path. The modification therefore places $\mathrm{s}(d_o)$ next to $\mathrm{t}(b_o)$ on the literal path of o for the first and the last occurrence o of each clause. For a clause with three literals, one of the three literal paths lies inside the cycle formed by the other two in every plane drawing of the gadget. In a drawing in which the anchors lie on the line as required, this inner path is the literal path of the second occurrence: the blocking targets of the first and the last occurrence lie on the line, and no point of the gadget lies between the line and the side of the box, so these targets lie on the outer face. The occurrence edge of the blocking token of the second occurrence, whose source lies outside the box, therefore has to cross one of the other two literal paths. The construction makes the crossing point an ordinary vertex of the gadget, which becomes the anchor of this occurrence edge, and it keeps the delay source of the second occurrence on the clause tail, which every route of the clause token uses. The proof of Lemma 18 gives the drawings of both gadgets. The construction is as follows.

Variable gadgets. The variable gadget of x is that of J, with the order of the vertices along its true and false paths fixed as follows. Along the false path, from g_x to $\mathrm{s}(a_x)$, the vertices $\mathrm{s}(b_o)$ and $\mathrm{t}(d_o)$ of the positive occurrences o of x appear in the order of these occurrences, with $\mathrm{s}(b_o)$ immediately followed by $\mathrm{t}(d_o)$ for each o, as in J. The order along the true path is defined symmetrically, with the negative occurrences of x in place of the positive ones.

Clause gadgets. The clause gadget of C has the vertices $\mathrm{s}(c_C)$, $\mathrm{t}(c_C)$, and g_C and, for each

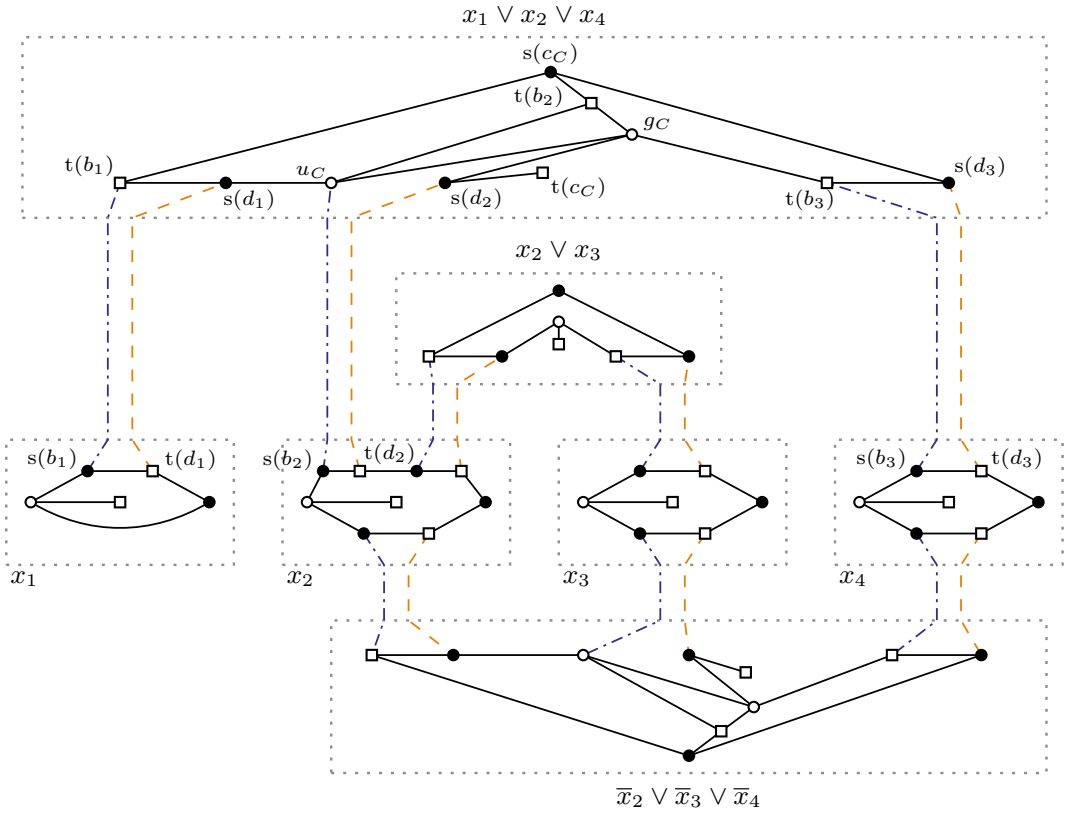


Figure 4 The graph G_{pl} for the formula of Figure 2, drawn as in the proof of Lemma 18 and restricted to the four variable gadgets and to the clause gadgets of $(x_1 \vee x_2 \vee x_4)$, $(x_2 \vee x_3)$, and $(\overline{x}_2 \vee \overline{x}_3 \vee \overline{x}_4)$; the gadgets of the other two clauses, their occurrence edges, and the vertices that their occurrences add to the variable gadgets are omitted. The dotted rectangles are the boxes of $\mathcal{R}$, and the dash-dotted blue curves and dashed orange curves are the occurrence edges of the blocking tokens and of the delay tokens, as in Figure 3. In the box of the clause $C = x_1 \vee x_2 \vee x_4$, whose occurrences o_1, o_2, and o_3 are those in x_1, x_2, and x_4, the vertices are labeled with the tokens indexed by occurrence, and so are the sources $\mathrm{s}(b_i)$ and targets $\mathrm{t}(d_i)$ of its occurrence edges in the boxes of x_1, x_2, and x_4; u_C is the crossing vertex of C, at which the occurrence edge of b_2 ends. In each variable gadget, the false path, which carries the vertices of the positive occurrences, is drawn above the variable line and the true path below it, whereas Figure 3 draws the true path above.

$o \in C$, the vertices $\mathrm{t}(b_o)$ and $\mathrm{s}(d_o)$, as in J, but its paths differ. If $|C| = 2$, the literal path of o_1 is $\mathrm{s}(c_C), \mathrm{t}(b_1), \mathrm{s}(d_1), g_C$, the literal path of o_2 is $\mathrm{s}(c_C), \mathrm{s}(d_2), \mathrm{t}(b_2), g_C$, and the clause tail is the edge $g_C\mathrm{t}(c_C)$; the gadget consists of the two literal paths and the clause tail. If $|C| = 3$, the gadget has one further ordinary vertex u_C, the *crossing vertex* of C (Figure 4). The literal path of o_1 is $\mathrm{s}(c_C), \mathrm{t}(b_1), \mathrm{s}(d_1), u_C, g_C$, the literal path of o_2 is $\mathrm{s}(c_C), \mathrm{t}(b_2), g_C$, the literal path of o_3 is $\mathrm{s}(c_C), \mathrm{s}(d_3), \mathrm{t}(b_3), g_C$, and the clause tail is $g_C, \mathrm{s}(d_2), \mathrm{t}(c_C)$; the gadget consists of the three literal paths, the clause tail, and the edge $u_C\mathrm{t}(b_2)$. Thus, u_C lies on the literal path of o_1, between $\mathrm{s}(d_1)$ and g_C, and is adjacent to $\mathrm{t}(b_2)$; it is the anchor of the occurrence edge of b_2, defined in the next item.

Occurrence edges. For each occurrence o, the occurrence edge of d_o is $\mathrm{s}(d_o)\mathrm{t}(d_o)$, as in J. The occurrence edge of b_o is $\mathrm{s}(b_o)\mathrm{t}(b_o)$, unless o is the second occurrence of a clause C with three literals, in which case it is $\mathrm{s}(b_o)u_C$. We also define, for each blocking or delay token z, its *occurrence path*, a path from $\mathrm{s}(z)$ to $\mathrm{t}(z)$. For every token z other

than the blocking token b_2 of a clause with three literals, the occurrence path of z is its occurrence edge alone. For the blocking token b_2 of a clause C with three literals, whose occurrence edge ends at u_C rather than at $\mathrm{t}(b_2)$, the occurrence path is $\mathrm{s}(b_2), u_C, \mathrm{t}(b_2)$, the occurrence edge followed by the edge $u_C\mathrm{t}(b_2)$ of the clause gadget.

Labels. The labels are those of J.

This completes the construction of J_{pl}. We first describe its basic properties.

▶ **Lemma 17.** *The instance J_{pl} can be computed in polynomial time, and its compatible matching is unique. The gadgets of G_{pl} are pairwise vertex-disjoint, and every edge of G_{pl} with endpoints in two gadgets is an occurrence edge, which joins the variable gadget and the clause gadget of its occurrence. The maximum degree of G_{pl} is four, and its vertices of degree four are the crossing vertices and the junctions of the clauses with three literals, all of which are ordinary.*

Proof. Compared with J, each clause with three literals contributes one extra vertex, its crossing vertex, and a constant number of extra edges, and the orders along the paths are obtained by sorting the ports of $\mathcal{R}$ from left to right, so J_{pl} can be computed in polynomial time. The labels are those of J, so the compatible matching remains unique. The gadgets of J are pairwise vertex-disjoint, and each crossing vertex is added to one clause gadget, so the gadgets of G_{pl} are pairwise vertex-disjoint as well. Every edge of G_{pl} is an edge of a gadget or an occurrence edge, and the occurrence edge of d_o joins $\mathrm{s}(d_o)$, in the clause gadget of o, to $\mathrm{t}(d_o)$, in the variable gadget of o, while that of b_o joins $\mathrm{s}(b_o)$, in the variable gadget of o, to $\mathrm{t}(b_o)$ or to the crossing vertex, in the clause gadget of o.

For the degrees, let C be a clause with three literals. The crossing vertex u_C is adjacent to $\mathrm{s}(d_1)$, g_C, and $\mathrm{t}(b_2)$ in the gadget and to $\mathrm{s}(b_2)$ by an occurrence edge, and g_C is adjacent to the last internal vertex of each of the three literal paths and to $\mathrm{s}(d_2)$; both have degree four, and both are ordinary. Each vertex of G_{pl} other than these has degree at most three. The vertices $\mathrm{s}(c_C)$, $\mathrm{t}(c_C)$, $\mathrm{s}(a_x)$, $\mathrm{t}(a_x)$, g_x, and the junction of a clause with two literals are incident with no occurrence edge and with at most three edges of their gadgets. The vertex $\mathrm{t}(b_2)$ of a clause C with three literals is incident with no occurrence edge and with exactly three edges of its gadget. Each remaining source or target of a blocking or delay token is incident with two edges of its gadget and one occurrence edge. ◀

We now show that G_{pl} is planar by giving the drawing described before the construction.

▶ **Lemma 18.** *The graph G_{pl} is planar.*

Proof. We draw each gadget inside its box so that, for each side of the box on which ports lie, the anchors of the occurrence edges arriving at that side lie on a horizontal line segment parallel to that side, in the left-to-right order of the ports of these edges, the anchor of the edge of b_o immediately followed by that of the edge of d_o, and that no point of the gadget lies strictly between the line through this segment and that side of the box. We first show that such drawings of the gadgets yield a plane drawing of G_{pl}, and then that they exist.

Outside the boxes, the segments of distinct occurrences are disjoint, and the segment of o meets only the two boxes of o, at the ports of o. We draw the occurrence edges of b_o and d_o between the two boxes as vertical line segments immediately to the left and to the right of the segment of o, so close to it that line segments of distinct occurrences are disjoint and each ends on the box sides next to the ports of o. Along each side, these endpoints thus occur in the order of the ports, that of b_o immediately before that of d_o, which is the order of the anchors on the horizontal line segment facing that side. We complete each occurrence

edge by a straight line segment from its endpoint on the side to its anchor. These joining line segments are pairwise disjoint, since they join points of two parallel lines in the same left-to-right order, and they meet the gadget only at the anchors, since every point of a joining line segment other than its anchor lies strictly between the line through the anchors and the side of the box, where the gadget has no point. Together with the pairwise disjointness of the boxes and of the vertical line segments, this makes the drawing of G_{pl} plane.

It remains to draw the gadgets. In the variable gadget of x, the anchors on the upper side are the vertices $\mathrm{s}(b_o)$ and $\mathrm{t}(d_o)$ of the positive occurrences o of x, which by construction appear along the false path in the required order, and those on the lower side are the corresponding vertices of the negative occurrences along the true path. We draw the false path as a polygonal line above the variable line with its internal vertices on a horizontal line segment, the true path symmetrically below, and the variable tail along the variable line between them; g_x and $\mathrm{s}(a_x)$ lie on the variable line, so no point of the gadget lies above the line through the first segment or below the line through the second. In the clause gadget of a positive clause C, the anchors are $\mathrm{t}(b_1), \mathrm{s}(d_1), \mathrm{t}(b_2), \mathrm{s}(d_2)$ if $|C| = 2$ and $\mathrm{t}(b_1), \mathrm{s}(d_1), u_C, \mathrm{s}(d_2), \mathrm{t}(b_3), \mathrm{s}(d_3)$ if $|C| = 3$, in the required order, since the occurrence edge of b_2 ends at u_C when $|C| = 3$. Figure 4 shows a plane drawing of each of the two gadgets with these vertices on a horizontal line segment in this order and no point of the gadget below the line through this segment: $\mathrm{s}(c_C)$ is joined to the two outermost anchors, g_C lies between them and is joined to its neighbors on the literal paths of o_1 and $o_{|C|}$, the clause tail lies below g_C, and, for $|C| = 3$, the literal path of o_2 runs from $\mathrm{s}(c_C)$ through $\mathrm{t}(b_2)$ to g_C with the edge $u_C\mathrm{t}(b_2)$ beside it. For a negative clause, we reflect this drawing across a horizontal line. ◀

We next show that the two properties of J stated above hold in J_{pl}.

▶ **Lemma 19.** *In every solution of J_{pl}, the route of each assignment token a_x consists of either the true path or the false path of its variable gadget, followed by its variable tail, and the route of each clause token c_C lies in the clause gadget of C, ends with the clause tail, and contains $\mathrm{t}(b_o)$ and $\mathrm{s}(d_o)$ for some occurrence $o \in C$.*

Proof. The proof of Observation 12 uses only that $\mathrm{s}(z)$ and $\mathrm{t}(z)$ are the source and the target of the token z alone, so the observation holds in J_{pl}: no route of a token other than z contains both $\mathrm{s}(z)$ and $\mathrm{t}(z)$. The occurrence edges fall into two classes. An occurrence edge of the form $\mathrm{s}(z)\mathrm{t}(z)$ is used by no route of an assignment token or of a clause token, by the observation. The remaining occurrence edges are the edges $\mathrm{s}(b_2)u_C$ of the clauses C with three literals, which the observation does not exclude, since u_C is not the target of b_2. Let G'_{pl} be obtained from G_{pl} by deleting the occurrence edges of the first class; every route of an assignment token or of a clause token is a path in G'_{pl}. By Lemma 17, the only edges of G'_{pl} joining two gadgets are the edges $\mathrm{s}(b_2)u_C$, and such an edge is the only edge of G'_{pl} with exactly one endpoint in the clause gadget of C, so it is a bridge of G'_{pl}, whose deletion separates the clause gadget of C from the rest of its component. A route of an assignment token has both endpoints in one variable gadget, and a route of a clause token has both endpoints in one clause gadget, so in both cases the two endpoints lie on the same side of each such bridge. A path that uses a bridge once ends on the other side of it, and a path that uses it twice is not simple; hence neither route uses an edge $\mathrm{s}(b_2)u_C$. Therefore, neither route uses an edge joining two gadgets, and each lies inside the gadget of its token.

In the variable gadget of x, the only edge incident with $\mathrm{t}(a_x)$ is the variable tail, so the route of a_x ends with the tail. The rest of the route is a simple path from $\mathrm{s}(a_x)$ to g_x that avoids $\mathrm{t}(a_x)$, that is, a simple path in the cycle formed by the true path and the false path.

The only such paths are the two arcs of this cycle between $s(a_x)$ and g_x, which are the true path and the false path.

In the clause gadget of C, the only edge incident with $t(c_C)$ is the last edge of the clause tail, and, if $|C| = 3$, the only neighbors of $s(d_2)$ in the gadget are g_C and $t(c_C)$. Hence, the route of c_C ends with the clause tail. The second vertex of the route is a neighbor of $s(c_C)$, that is, $t(b_1)$, $s(d_{|C|})$, or, if $|C| = 3$, $t(b_2)$. If it is $t(b_1)$, then the third vertex is $s(d_1)$, the only other neighbor of $t(b_1)$ in the gadget, so the route contains $t(b_1)$ and $s(d_1)$. If it is $s(d_{|C|})$, then the third vertex is $t(b_{|C|})$, the only other neighbor of $s(d_{|C|})$ in the gadget, so the route contains $t(b_{|C|})$ and $s(d_{|C|})$. If it is $t(b_2)$, then the route contains $t(b_2)$ and, on the clause tail, $s(d_2)$. ◀

As for J, a solution of J_{pl} with route family Π determines the truth assignment τ_Π that sets x to `true` if a_x uses its true path and to `false` if a_x uses its false path; by Lemma 19, τ_Π is well defined. In J, every route of a clause token c_C contains the delay source of every occurrence in C, and this is what forces the blocking token of every false occurrence in C to move before c_C (Observation 15). In J_{pl}, the route of c_C contains only some of these delay sources, so the next lemma establishes this precedence only for the occurrences whose delay source lies on the route of c_C.

▶ **Lemma 20.** *Let o be an occurrence in a clause C, and let x be the variable of o. In every solution of J_{pl} with route family Π in which o is false under τ_Π and the route of c_C contains $s(d_o)$, the token b_o moves before a_x, a_x moves before d_o, d_o moves before c_C, and $t(b_o)$ is occupied when c_C moves.*

Proof. By construction, $s(b_o)$ and $t(d_o)$ lie on the false path of the variable gadget of x if o is positive and on its true path if o is negative. Since o is false under τ_Π, both vertices lie on the route of a_x by Lemma 19. The source $s(b_o)$ must be vacated before a_x moves, so b_o moves before a_x. Moving d_o before a_x would occupy $t(d_o)$, an internal vertex of the route of a_x, so a_x moves before d_o. The source $s(d_o)$ is an internal vertex of the route of c_C, so d_o moves before c_C. Hence b_o moves before c_C, and since each token moves at most once, $t(b_o)$ is occupied when c_C moves. ◀

▶ **Theorem 2.** *Single-Move Labeled Token Routing on unique-matching instances is NP-complete on planar graphs of maximum degree four, even on instances that, when solvable, admit a solution with pairwise edge-disjoint routes, and on planar graphs of maximum degree three, even on instances that, when solvable, admit a solution with solution edge multiplicity at most two.*

Proof. By Lemma 17, J_{pl} is computed in polynomial time, is a unique-matching instance, and has maximum degree four, and by Lemma 18 its graph is planar. We first show that φ is satisfiable if and only if J_{pl} is a yes-instance.

($\Rightarrow$) Suppose that φ has a satisfying assignment τ. We move the tokens in the five stages of the proof of Lemma 16: first the blocking tokens of the occurrences that are false under τ, then the assignment tokens, the delay tokens, the clause tokens, and finally the remaining blocking tokens. Every blocking or delay token moves along its occurrence path, a_x moves along its true path if $\tau(x) =$ `true` and along its false path otherwise, followed by its variable tail, and c_C moves along the literal path of an occurrence in C that is true under τ, followed by the clause tail. The occurrence path of a blocking or delay token has no internal vertex other than, possibly, a crossing vertex, which is ordinary, and its target is neither a source nor the target of another token, so the route is clear when the token moves. The internal

vertices of the route of a_x are g_x and the vertices $\mathrm{s}(b_o)$ and $\mathrm{t}(d_o)$ of the occurrences o of x on the path along which a_x moves, which are the occurrences of x that are false under τ; their blocking tokens moved in the first stage, their delay tokens have not yet moved, and $\mathrm{t}(a_x)$ is unoccupied, so the route is clear. The internal vertices of the route of c_C are g_C, possibly u_C, the vertex $\mathrm{t}(b_o)$ of the occurrence o whose literal path c_C uses, and delay sources; b_o has not moved because o is true, every delay token has moved, and $\mathrm{t}(c_C)$ is unoccupied, so the route is clear. Hence, J_{pl} is a yes-instance.

($\Leftarrow$) Suppose that J_{pl} is a yes-instance, let Π be the route family of a solution, and consider a clause C. By Lemma 19, the route of c_C contains $\mathrm{t}(b_o)$ and $\mathrm{s}(d_o)$ for some occurrence $o \in C$. If o were false under τ_Π, then by Lemma 20 $\mathrm{t}(b_o)$ would be occupied when c_C moves, and the route of c_C would not be clear. Hence, o is true under τ_Π, so τ_Π satisfies C. Since this holds for every clause, τ_Π satisfies φ.

Bounding the maximum degree. By Lemma 17, the vertices of degree four of G_{pl} are the crossing vertices and the junctions of the clauses with three literals, and all of them are ordinary. We apply Lemma 11 to each of them, expanding it by a tree with four leaves and two internal vertices. The resulting instance is a yes-instance if and only if J_{pl} is, its graph is planar and has maximum degree three, and its compatible matching remains unique, since only ordinary vertices are introduced.

Bounding the solution edge multiplicity. Consider the solution constructed from a satisfying assignment in the forward direction. The routes of the assignment tokens and of the clause tokens lie in pairwise vertex-disjoint gadgets and use only edges of true paths, false paths, variable tails, literal paths, and clause tails. The occurrence path of a blocking or delay token consists of its occurrence edge and, for the second occurrence of a clause C with three literals, the edge $u_C\mathrm{t}(b_2)$, which lies on no literal path and on no clause tail. Hence, the routes of this solution are pairwise edge-disjoint, so every constructed instance that is a yes-instance admits a solution with solution edge multiplicity one.

For the instance of maximum degree three obtained by the degree reduction, modify the solution above as follows: every route that passes through an expanded vertex now passes through the tree that replaced it, entering it through the leaf adjacent to the preceding vertex of the route and leaving it through the leaf adjacent to the following one. The vertices of the tree are ordinary, so these routes are clear whenever the original ones were, and the result is a solution. At a crossing vertex u_C, the routes of b_2 and of c_C may both pass, entering and leaving the tree through four distinct leaves, so they share at most the edge joining the two internal vertices of the tree; at an expanded junction only the clause token passes. Hence, every constructed instance of maximum degree three that is a yes-instance admits a solution with solution edge multiplicity at most two. Together with membership in NP, established in Section 4, this proves the theorem. ◀

5.2.3 Grids and grid graphs

To prove Theorem 3, we construct, from the planar instance J_{pl} of Section 5.2.2, an instance whose graph is an induced subgraph of a grid. A *grid point* is a point of the plane with integer coordinates, so that the vertices of a grid are grid points and the grid with vertex set the grid points of an axis-parallel rectangle is the grid *induced* by that rectangle; a *grid line* is a horizontal or vertical line through grid points. Since G_{pl} is planar and has maximum degree four, it admits an orthogonal grid drawing, in which every vertex is a grid point and every edge is drawn as a path along grid lines. The grid points on such a drawing induce a subdivision of G_{pl}, since every bend and every intermediate grid point on the drawing of an

edge becomes a vertex. We therefore first show that subdividing edges preserves feasibility.

▶ **Observation 21.** *Let $I = (G, S, T, L)$ be an instance of Single-Move Labeled Token Routing, and let G' be obtained from G by subdividing any number of edges, every subdivision vertex being an ordinary vertex of the instance $I' = (G', S, T, L)$. Then I is a yes-instance if and only if I' is a yes-instance.*

Proof. Suppose first that I is a yes-instance, and let M, Π, and π be a compatible matching, route family, and feasible movement order, respectively, that constitute a solution. Replace every edge of every route in Π by the corresponding subdivided path in G'. Since every subdivision vertex is ordinary, it is permanently unoccupied. Hence, π is feasible for the resulting route family, and I' is a yes-instance.

Conversely, suppose that I' is a yes-instance. Contract every subdivided path to the corresponding edge of G. This transforms every route into a simple path with the same endpoints. At the time a token moves, every internal vertex of the resulting route was also an internal vertex of its route in G' and is therefore unoccupied. Hence, the resulting matching, route family, and movement order form a solution of I. ◀

▶ **Theorem 3.** *Single-Move Labeled Token Routing on unique-matching instances is NP-complete on grids and on grid graphs, even on instances that, when solvable, admit a solution with pairwise edge-disjoint routes.*

Proof. For hardness on grid graphs, consider the planar instance $J_{\mathrm{pl}} = (G_{\mathrm{pl}}, S, T, L)$ of Section 5.2.2. We use G_{pl} rather than the graph of maximum degree three obtained by the degree reduction, since we need a solution with pairwise edge-disjoint routes. Compute an orthogonal grid drawing of G_{pl} of polynomial size, and multiply all coordinates by two, so that two grid points of the drawing are at distance one only if they are consecutive on the drawing of one edge. Let $G_{\square}$ be the graph whose vertices are the grid points lying on the drawing and whose edges join consecutive grid points along the drawing of an edge of G_{pl}.

By construction, $G_{\square}$ is a subdivision of G_{pl} of polynomial size. It is also an induced subgraph of the grid: a grid edge joining two vertices of $G_{\square}$ joins two grid points of the drawing at distance one, which by the doubling are consecutive on the drawing of one edge, so the grid edge is an edge of $G_{\square}$.

By Observation 21, $(G_{\mathrm{pl}}, S, T, L)$ is a yes-instance if and only if $(G_{\square}, S, T, L)$ is a yes-instance. Since the labels are unchanged, every source in $G_{\square}$ remains compatible with exactly one target. If $(G_{\square}, S, T, L)$ is a yes-instance, then so is $(G_{\mathrm{pl}}, S, T, L)$, which therefore admits a solution with pairwise edge-disjoint routes; replacing every edge of each of its routes by the path of $G_{\square}$ that subdivides it yields a solution of $(G_{\square}, S, T, L)$ whose routes are again pairwise edge-disjoint, since the paths that subdivide distinct edges are edge-disjoint. This proves NP-hardness on grid graphs.

To obtain hardness on grids, let H be the grid induced by the smallest axis-parallel rectangle containing the drawing. For every vertex $u \in V(H) \setminus V(G_{\square})$, make u both the source and the target of an additional token, and assign a fresh private label to u as a source and as a target. The only simple route of this token is the one-vertex path at u, so u remains occupied throughout every movement order.

Consequently, no token of the instance on $G_{\square}$ can use a vertex of $H \setminus V(G_{\square})$ as an internal vertex of its route. Moreover, by the doubling, H contains no edge with both endpoints in $V(G_{\square})$ that does not already belong to $G_{\square}$. Therefore, adding the vertices and edges of $H \setminus G_{\square}$ creates no new route for any original token.

A solution for the instance on $G_{\square}$ extends to the padded instance on H by matching each added source with itself, using the one-vertex route. Conversely, removing these pairs

from a solution on H yields a solution on $G_\square$. The added tokens have private labels, and the original sources retain their unique compatible targets. The added routes use no edge, so pairwise edge-disjoint routes remain pairwise edge-disjoint. Hence, Single-Move Labeled Token Routing on unique-matching instances is NP-complete on grids, even on instances that, when solvable, admit a solution with pairwise edge-disjoint routes. ◀

5.3 Solution edge multiplicity on trees of maximum degree three

We next turn to trees. On a tree, every compatible source-target pair is connected by a unique path, so there is no choice of routes. Thus, any hardness on trees must arise from the interaction between choosing a compatible matching and ordering the resulting moves. The result of Geft [3] recalled in Section 2 shows that these choices alone suffice to make Single-Move Labeled Token Routing NP-hard on instances whose tree has maximum degree three.

We further ask whether the complexity of such instances can be explained by the amount of overlap among the routes of a solution. Our next result shows that Single-Move Labeled Token Routing is W[1]-hard when parameterized by an upper bound μ, given as part of the input, on the solution edge multiplicity, even on trees of maximum degree three, and even when a movement order is given as part of the input.

We prove the result by a parameterized reduction from Grid Tiling, in which we view an ordered pair $(i, j) \in [K]^2$ as a *cell* of a $K \times K$ grid, to be filled with a *tile*, that is, a pair of values in $[N]^2$. Formally, an instance of Grid Tiling consists of two positive integers K and N together with a set $\mathcal{T}_{i,j} \subseteq [N]^2$ for each cell $(i, j) \in [K]^2$; the tiles in $\mathcal{T}_{i,j}$ are the *legal tiles for cell* (i, j).

A solution is an assignment of a value in $[N]$ to each row and to each column such that the two values assigned to the row and the column of a cell form a legal tile for it, that is, if a is assigned to row i and b is assigned to column j, then $(a, b) \in \mathcal{T}_{i,j}$. The problem is NP-hard, and W[1]-hard when parameterized by K [44, 47].

To encode the Grid Tiling instance, we construct an instance of Single-Move Labeled Token Routing whose tree consists of a *tile subtree*, $2K$ *choice subtrees*, one for each row and one for each column, two *storage paths*, and a *cleanup path*, connected as described below and shown in Figure 5. The choice subtrees are used to choose a value for each row and each column, while the tile subtree is used to verify that the chosen values specify a legal tile for each cell. Each legal tile $(a, b) \in \mathcal{T}_{i,j}$ is associated with two *tile tokens*: a *row tile token*, initially stored in the choice subtree for row i, and a *column tile token*, initially stored in the choice subtree for column j. Choosing the tile (a, b) for cell (i, j) consists of moving its column tile token to the tile subtree, and its row tile token into the target that the column tile token leaves free. The choice subtree of a row or column contains one *sealing token* for each value, and moving it to a storage path, which contains targets for the sealing tokens, chooses that value. The cleanup path contains the sources of *cleanup tokens*, used to fill targets in the choice subtrees.

Tile subtree. The tile subtree is a star with K^2 leaves, each a target vertex with a distinct label set $\{\text{tile}(i, j)\}$ for $(i, j) \in [K]^2$; the target with label set $\{\text{tile}(i, j)\}$ is the *target for cell* (i, j). The sources whose label sets contain $\text{tile}(i, j)$ are the column tile token sources, introduced below.

Choice subtrees. We construct $2K$ choice subtrees, one for each row and one for each column. Each choice subtree contains N *value subtrees*, one for each of the N possible values. We denote the choice subtree for row i by Row_i and its value subtree for value a by $\text{Row}_i(a)$.

Similarly, we denote the choice subtree for column j by Col_j and its value subtree for value b by $\mathrm{Col}_j(b)$.

The construction must ensure that the tile tokens that leave a choice subtree are those of a single value subtree, which then represents the value chosen for that row or column. To that end, each value subtree is connected to the rest of the tree through a single vertex, its *attachment vertex*, which is both a source and a target. At the attachment vertex of $\mathrm{Row}_i(a)$, the label set of the source and the label set of the target are both $\{\mathrm{seal}(\mathrm{Row}_i(a))\}$, and at the attachment vertex of $\mathrm{Col}_j(b)$, they are both $\{\mathrm{seal}(\mathrm{Col}_j(b))\}$. The token initially placed on an attachment vertex is the *sealing token* of its value subtree. Since the source and the target at an attachment vertex share their label, a solution has two ways of matching a sealing token. Leaving it at its own source keeps the attachment vertex occupied, so that no tile token can leave the value subtree and each tile token must instead move to a *local target* inside the subtree; we then call the value subtree *sealed*. Matching the sealing token with a target outside the choice subtree instead frees the attachment vertex; we call the value subtree *unsealed*. The storage paths described below supply only one such outside target per choice subtree, so at most one value subtree per choice subtree can be unsealed.

The remaining vertices of a value subtree form a path of *tile token sources*, one end of which is adjacent to the attachment vertex, together with one *local target* vertex for each tile token source, adjacent to it as a leaf. In $\mathrm{Row}_i(a)$, there is one tile token source for each $j \in [K]$ and each $b \in [N]$ with $(a, b) \in \mathcal{T}_{i,j}$, and the token initially placed on the source for j and b is the row tile token for (a, b). In $\mathrm{Col}_j(b)$, there is one tile token source for each $i \in [K]$ and each $a \in [N]$ with $(a, b) \in \mathcal{T}_{i,j}$, and the token initially placed on the source for i and a is the column tile token for (a, b).

Because of the differing roles of column tile tokens and row tile tokens, the labels of vertices in value subtrees differ depending on whether they belong to a choice subtree for a column or for a row. In $\mathrm{Col}_j(b)$, the tile token source for i and a has label set $\{\mathrm{replacement}(i, j, a, b), \mathrm{tile}(i, j)\}$ and the local target has label set $\{\mathrm{replacement}(i, j, a, b)\}$. Thus, a column tile token can move either to the target for its cell in the tile subtree or to its own local target. Moreover, the only sources compatible with the local target of the column tile token for (a, b) are the source of that token itself and the source of the row tile token for (a, b) in $\mathrm{Row}_i(a)$.

In $\mathrm{Row}_i(a)$, the tile token source for j and b has label set $\{\mathrm{replacement}(i, j, a, b), \mathrm{clean}(i, j)\}$ and its local target has label set $\{\mathrm{clean}(i, j)\}$. We will use these labels to show that a row tile token moves either to the local target of the column tile token for the same tile, or to its own local target.

Storage paths. The *row-storage path* contains K target vertices. For every $i \in [K]$, the ith target has label set $\{\mathrm{seal}(\mathrm{Row}_i(a)) : a \in [N]\}$ and is therefore compatible with every sealing token in the choice subtree Row_i, but with no sealing token from any other row choice subtree.

Similarly, the *column-storage path* contains K target vertices. For every $j \in [K]$, the jth target has label set $\{\mathrm{seal}(\mathrm{Col}_j(b)) : b \in [N]\}$ and is therefore compatible with every sealing token in the choice subtree Col_j, but with no sealing token from any other column choice subtree.

Cleanup path. The *cleanup path* contains the following source vertices:

1. for every $(i, j) \in [K]^2$, one source with label set $\{\mathrm{clean}(i, j)\}$, the *cell source* for cell (i, j); it can be used to occupy the local target left unoccupied when the corresponding row tile token is moved to a choice subtree for a column;
2. for every $i \in [K]$, the ith *row source* has label set $\{\mathrm{seal}(\mathrm{Row}_i(a)) : a \in [N]\}$; it can be

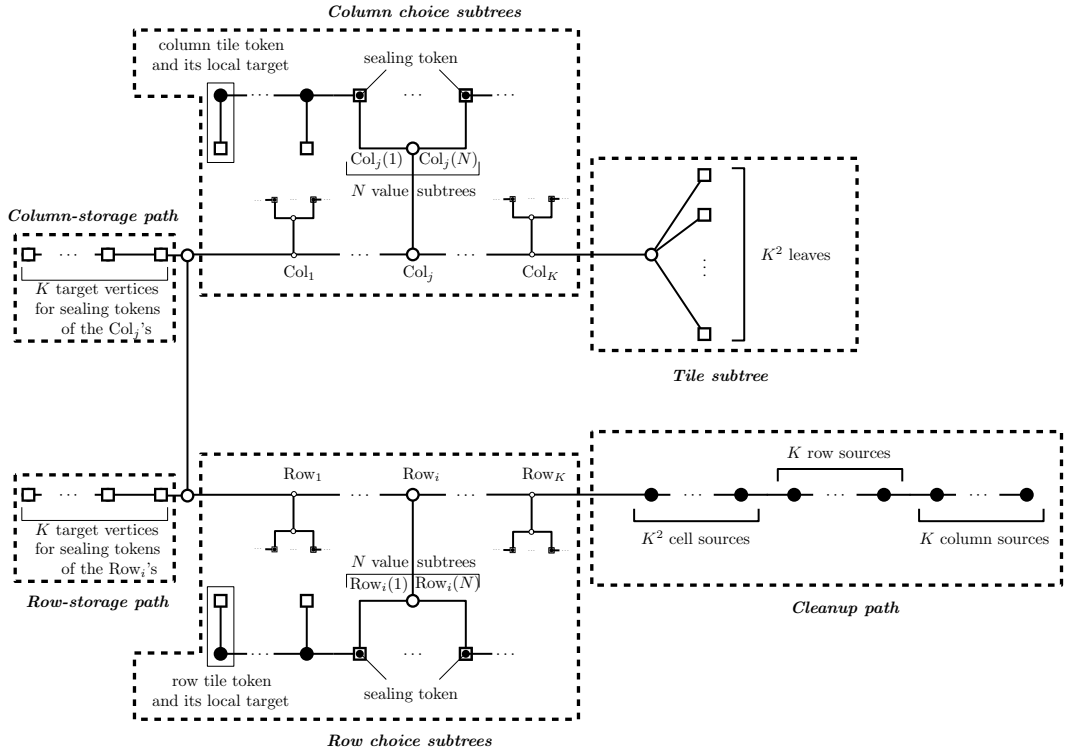


Figure 5 The arrangement of the choice subtrees, the tile subtree, the two storage paths, and the cleanup path in the instance constructed from a Grid Tiling instance.

used to replace a sealing token that moves from Row_i to the row-storage path;

3. for every $j \in [K]$, the jth *column source* has label set $\{\mathrm{seal}(\mathrm{Col}_j(b)) : b \in [N]\}$; it can be used to replace a sealing token that moves from Col_j to the column-storage path.

Connections. All vertices joining these parts are ordinary. In each choice subtree, the attachment vertices of the N value subtrees are adjacent to a common ordinary vertex. For the row choice subtrees $\mathrm{Row}_1, \ldots, \mathrm{Row}_K$, these common vertices are adjacent, in this order, to distinct vertices of a path of ordinary vertices, and for the column choice subtrees $\mathrm{Col}_1, \ldots, \mathrm{Col}_K$, to distinct vertices of a second such path. One end of the first path is adjacent to one end of the second path; the former end is also adjacent to an end of the row-storage path, and the latter to an end of the column-storage path. The other end of the second path is adjacent to the center of the tile subtree, and the other end of the first path is adjacent to an end of the cleanup path. Along the cleanup path, starting from that end, the K^2 cell sources come first, followed by the K row sources and then the K column sources.

In the constructed tree, the vertices of degree greater than three are the center of the tile subtree, the common neighbors of the attachment vertices in the choice subtrees, and the vertices at which the storage paths and the cleanup path are attached; all of them are ordinary, since every source and every target of the construction lies inside a value subtree, is a leaf of the tile subtree, or lies on a storage path or on the cleanup path. We may therefore apply the operation of Lemma 11 to each of them and obtain, in polynomial time, an equivalent instance in which the tree has maximum degree three.

For ease of reference, we summarize which sources and targets use each label, as well as

their uses:

Tile labels. The label $\text{tile}(i, j)$ is associated with cell (i, j). It appears as a source label for the column tile token for (a, b) for each legal tile (a, b) for cell (i, j), and as the target label of a single vertex in the tile subtree.

Sealing labels. The label $\text{seal}(\text{Row}_i(a))$ determines whether the value subtree $\text{Row}_i(a)$ is sealed. It labels a vertex in $\text{Row}_i(a)$ that is both source and target, the ith row source in the cleanup path, and the ith target in the row-storage path. Similarly, $\text{seal}(\text{Col}_j(b))$ labels a vertex in the value subtree $\text{Col}_j(b)$ that is both source and target, the jth column source in the cleanup path, and the jth target in the column-storage path.

Replacement labels. The label $\text{replacement}(i, j, a, b)$, for each cell $(i, j) \in [K]^2$ and each tile $(a, b) \in \mathcal{T}_{i,j}$, appears in the label sets of the tile token source and of the local target in $\text{Col}_j(b)$ and of the corresponding tile token source in $\text{Row}_i(a)$. Thus, either the tile token in $\text{Col}_j(b)$ or the tile token in $\text{Row}_i(a)$ can move to the local target in $\text{Col}_j(b)$.

Cleanup labels. The label $\text{clean}(i, j)$ labels a source in the cleanup path and, for every $a, b \in [N]$ with $(a, b) \in \mathcal{T}_{i,j}$, the tile token source for j and b in $\text{Row}_i(a)$ together with its local target.

Below we present two properties of the construction that are used in the proof.

▶ **Observation 22.** *Every route from a tile token source in a value subtree to a target outside that value subtree contains the source of the sealing token of that value subtree as an internal vertex. Moreover, every local target and every target for a cell in the tile subtree is a leaf of the constructed tree.*

Proof. By construction, the only edge joining a value subtree to the rest of the tree is incident with the vertex that is both the source and the target of its sealing token, and that vertex is not a tile token source; hence every path leaving the value subtree passes through it, and it is an internal vertex of such a path. The second statement holds because each local target is introduced as a leaf neighbor of a tile token source and each target for a cell is introduced as a leaf of the tile subtree. ◀

As mentioned above, a value subtree is *unsealed* if its sealing token is matched with the corresponding target on a storage path, and *sealed* otherwise; by the summary of sealing labels, the sealing token of a sealed value subtree is matched with the target at its own source.

Throughout this subsection, the given instance of Grid Tiling is the one fixed above, with parameters K and N and legal-tile sets $\mathcal{T}_{i,j}$, and J_{GT} denotes the constructed instance of Single-Move Labeled Token Routing after the operation of Lemma 11 has been applied to its vertices of degree greater than three, so that its tree has maximum degree three.

▶ **Lemma 23.** *The constructed Single-Move Labeled Token Routing instance can be computed in polynomial time. Moreover, if the Grid Tiling instance admits a solution, then J_{GT} admits a solution with solution edge multiplicity at most $3K^2 + 4K$.*

Proof. The construction contains $\mathcal{O}\big(KN + \sum_{(i,j)\in[K]^2} |\mathcal{T}_{i,j}| + K^2\big)$ vertices and edges. The exhaustive applications of Lemma 11 increase the size by at most a polynomial factor. Hence, the constructed instance can be computed in polynomial time.

Suppose that the Grid Tiling instance admits a solution. Let $a_1, \ldots, a_K \in [N]$ and $b_1, \ldots, b_K \in [N]$ satisfy $(a_i, b_j) \in \mathcal{T}_{i,j}$ for every $(i, j) \in [K]^2$. We construct a compatible

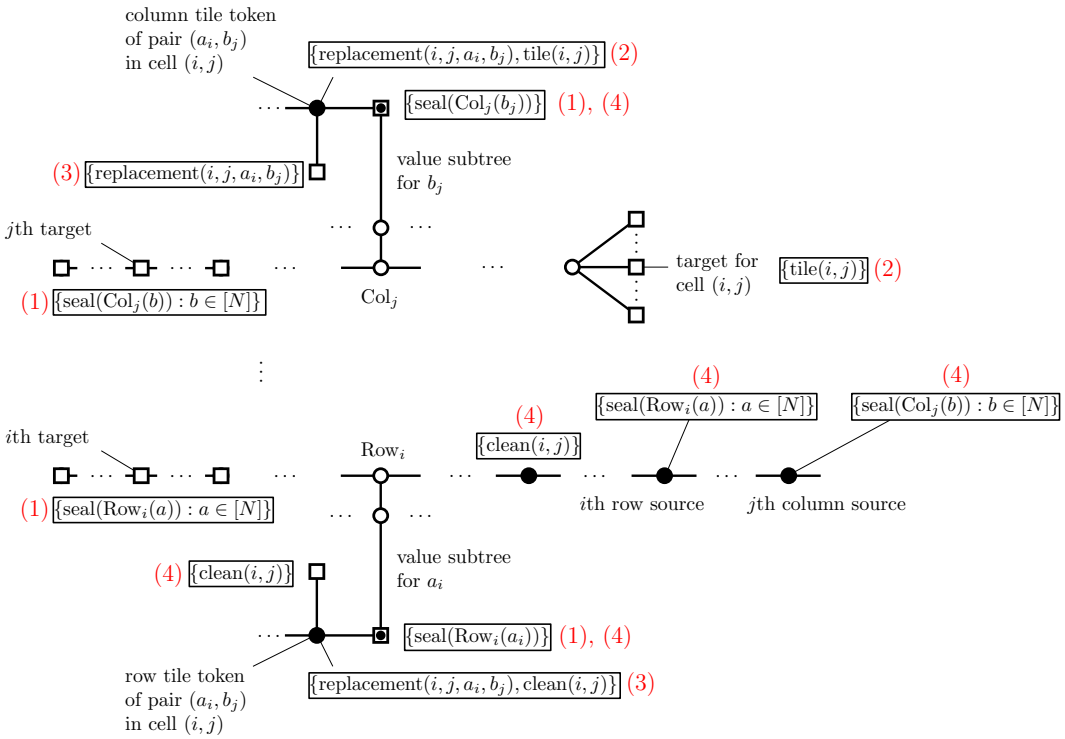


Figure 6 The movement order associated with a fixed cell (i, j), assuming that the value a_i is chosen for row i and the value b_j for column j. The numbers in parentheses identify the order in which the moves are to be executed, and their positions indicate the source and the target of that move. The label sets of the indicated sources and targets are displayed inside rectangles. Ellipses represent portions of the construction not relevant to the depicted moves: the gadgets retain their relative positions from Figure 5.

matching, use the unique tree path for each matched pair as its route, and specify a feasible movement order. The movement order is depicted in Figure 6.

1. **Sealing tokens ((1) in Figure 6).** For every row $i \in [K]$, match the sealing token in $\text{Row}_i(a_i)$ with the ith target of the row-storage path. For every $a \in [N] \setminus \{a_i\}$, match the sealing token in $\text{Row}_i(a)$ with the target at its own source. Thus, $\text{Row}_i(a_i)$ is unsealed, whereas all other value subtrees of Row_i remain sealed.
   Similarly, for every column $j \in [K]$, match the sealing token in $\text{Col}_j(b_j)$ with the jth target of the column-storage path, and match all other sealing tokens in Col_j with the targets at their own source. Thus, $\text{Col}_j(b_j)$ is unsealed, whereas all other value subtrees of Col_j remain sealed.
   Move the sealing tokens matched with storage targets before moving any tile token, and move those matched with the same storage path in order of decreasing distance of their storage targets from the vertex at which that path is attached to the rest of the tree. Then, when a sealing token moves, no storage target on its route is occupied.
2. **Column tile tokens ((2) in Figure 6).** For every cell $(i, j) \in [K]^2$, match the column tile token for (a_i, b_j) in $\text{Col}_j(b_j)$ with the target for cell (i, j) in the tile subtree. These vertices are compatible through the label $\text{tile}(i, j)$.
   Match all other column tile tokens with their local targets. Thus, the local target of the column tile token for (a_i, b_j) remains unoccupied, whereas all other local targets of

column tile tokens are occupied by the corresponding tile tokens.

In each column value subtree, move the column tile tokens from closest to farthest from the source of the value subtree's sealing token. The routes of the column tile tokens moved to the tile subtree are clear because the relevant value subtrees are unsealed and every tile token source closer to the sealing token's source has already been vacated. Since the targets in the tile subtree and the local targets of column tile tokens are leaves (Observation 22), occupying one of these targets does not block a later route of a column tile token.

3. **Row tile tokens ((3) in Figure 6).** For every cell $(i, j) \in [K]^2$, match the row tile token for (a_i, b_j) in $\mathrm{Row}_i(a_i)$ with the local target of the column tile token for (a_i, b_j) in $\mathrm{Col}_j(b_j)$. These vertices are compatible through the label replacement(i, j, a_i, b_j).

   Match all other row tile tokens with their local targets. Thus, for every cell (i, j), the row tile token for (a_i, b_j) occupies the local target left unoccupied by the corresponding column tile token.

   After all column tile tokens have moved, move the row tile tokens in every row value subtree from closest to farthest from its sealing token's source. The routes to local targets in column choice subtrees are clear: the value subtrees $\mathrm{Row}_i(a_i)$ and $\mathrm{Col}_j(b_j)$ are unsealed, every row tile token source closer to the sealing token's source has already been vacated, and so have all column tile token sources.

4. **Cleanup tokens ((4) in Figure 6).** For every cell $(i, j) \in [K]^2$, the local target of the row tile token for (a_i, b_j) remains unoccupied. Match the cleanup token with label clean(i, j) with this target.

   For every row $i \in [K]$, match the ith row source of the cleanup path with the target at the source of the sealing token in $\mathrm{Row}_i(a_i)$. Similarly, for every column $j \in [K]$, match the jth column source of the cleanup path with the target at the source of the sealing token in $\mathrm{Col}_j(b_j)$.

   First move the cleanup tokens with labels of the form clean(i, j), from closest to farthest from the vertex at which the cleanup path is attached to the rest of the tree. At this point, every row tile token source on the corresponding route has been vacated. Next, move the remaining cleanup tokens, again from closest to farthest from that vertex. Since all tile tokens have already moved, the routes to the sources of the sealing tokens of the unsealed value subtrees are clear.

Every source and every target occurs exactly once in the proposed matching. Every sealing token in a sealed value subtree is matched with the target at its own source, while every sealing token in an unsealed value subtree is replaced by the appropriate cleanup token. Every target for a cell in the tile subtree is matched with one column tile token. For every cell, the local target of this column tile token is matched with the corresponding row tile token, and the local target of that row tile token is matched with the cleanup token carrying the corresponding cleanup label. Every remaining tile token is matched with its local target.

All nontrivial routes are clear when they are used. Therefore, the matching, the unique tree paths between its matched pairs, and the movement order form a solution of J_{GT}.

The only routes that can use an edge shared with another route are the K^2 routes from column choice subtrees to the tile subtree, the K^2 routes from row choice subtrees to local targets in column choice subtrees, the K^2 routes from the cleanup path to local targets in row choice subtrees, the $2K$ routes from unsealed value subtrees to the storage paths, and the $2K$ routes from the cleanup path to the sources of the sealing tokens in unsealed value subtrees. Every remaining route is trivial or consists of the single edge joining a tile token source to its local target; since that local target is a leaf matched with no other source, no

other route uses this edge. Hence, every edge, including every edge of a tree introduced by the operation of Lemma 11, is used by at most $3K^2 + 4K$ routes, so this solution has solution edge multiplicity at most $3K^2 + 4K$. ◀

We next establish the properties of an arbitrary solution of J_{GT} that are needed for the reverse direction.

▶ **Lemma 24.** *For every row $i \in [K]$, at most one value subtree of Row_i is unsealed. Similarly, for every column $j \in [K]$, at most one value subtree of Col_j is unsealed.*

Proof. Fix a row $i \in [K]$. As mentioned in the summary of sealing labels, every sealing token in Row_i that is not matched with the target at its own source must be matched with the ith target of the row-storage path. Since a matching uses this target at most once, at most one sealing token in Row_i can be matched with the row-storage path. Hence, at most one value subtree of Row_i is unsealed.

The argument for a column choice subtree is symmetric. ◀

▶ **Lemma 25.** *For every cell $(i, j) \in [K]^2$, there exists a legal tile $(a, b) \in \mathcal{T}_{i,j}$ such that the following hold:*

1. *the target for cell (i, j) in the tile subtree is matched with the source of the column tile token for (a, b) in $\mathrm{Col}_j(b)$;*
2. *this column tile token is moved from $\mathrm{Col}_j(b)$ to the target for cell (i, j) in the tile subtree;*
3. *the local target of this column tile token is matched with the source of the row tile token for (a, b) in $\mathrm{Row}_i(a)$; and*
4. *this row tile token is moved from $\mathrm{Row}_i(a)$ to the local target of the column tile token.*

Consequently, $\mathrm{Col}_j(b)$ and $\mathrm{Row}_i(a)$ are unsealed.

Proof. Fix a cell $(i, j) \in [K]^2$. As discussed in the summary of tile labels, the only sources compatible with the target for cell (i, j) in the tile subtree are the sources of column tile tokens for legal tiles of cell (i, j). Since every target is matched, this target is matched with the source of a column tile token for some legal tile $(a, b) \in \mathcal{T}_{i,j}$ in $\mathrm{Col}_j(b)$. This proves the first statement.

The target for cell (i, j) lies outside $\mathrm{Col}_j(b)$, so the corresponding column tile token is moved from $\mathrm{Col}_j(b)$ to that target, which proves the second statement. By Observation 22, its route contains the source of the sealing token of $\mathrm{Col}_j(b)$ as an internal vertex, so that source cannot remain occupied when the column tile token moves. Hence, $\mathrm{Col}_j(b)$ is unsealed.

As noted in the summary of replacement labels, the local target of this column tile token is compatible only with the source of the column tile token itself and with the source of the corresponding row tile token for (a, b) in $\mathrm{Row}_i(a)$. The column tile token is already matched with the target for cell (i, j) in the tile subtree and therefore cannot be matched with its local target. Hence, this local target is matched with the source of the corresponding row tile token. This proves the third statement.

Finally, the local target of the column tile token lies outside $\mathrm{Row}_i(a)$. Thus, the corresponding row tile token is moved from $\mathrm{Row}_i(a)$ to this target, which proves the fourth statement. By Observation 22, its route contains the source of the sealing token of $\mathrm{Row}_i(a)$ as an internal vertex, so that source cannot remain occupied when the row tile token moves. Hence, $\mathrm{Row}_i(a)$ is unsealed. ◀

▶ **Theorem 4.** *On trees of maximum degree three, Single-Move Labeled Token Routing is NP-complete, and, for a nonnegative integer μ given as part of the input, deciding*

whether there is a solution with solution edge multiplicity at most μ is W[1]-hard when parameterized by μ; both results hold even when a movement order is given as part of the input.

Proof. By Lemma 23, the mapping from the Grid Tiling instance to J_{GT} can be computed in polynomial time. We show that the former is a yes-instance if and only if the latter is.

($\Rightarrow$) Suppose that the Grid Tiling instance is a yes-instance. Lemma 23 yields a solution of J_{GT}. Hence, J_{GT} is a yes-instance.

($\Leftarrow$) Suppose that J_{GT} is a yes-instance.

For every row $i \in [K]$, apply Lemma 25 to cell $(i, 1)$. This yields a value $a \in [N]$ such that $\mathrm{Row}_i(a)$ is unsealed. By Lemma 24, no other value subtree of Row_i is unsealed. Write a_i for this value.

Similarly, for every column $j \in [K]$, apply Lemma 25 to cell $(1, j)$. This yields a value $b \in [N]$ such that $\mathrm{Col}_j(b)$ is unsealed. By Lemma 24, no other value subtree of Col_j is unsealed. Write b_j for this value.

Now fix a cell $(i, j) \in [K]^2$. By Lemma 25, there exists a legal tile $(a, b) \in \mathcal{T}_{i,j}$ such that $\mathrm{Col}_j(b)$ and $\mathrm{Row}_i(a)$ are unsealed. By the definitions of a_i and b_j, together with Lemma 24, we have $a = a_i$ and $b = b_j$. Therefore, $(a_i, b_j) = (a, b) \in \mathcal{T}_{i,j}$.

Since this holds for every cell $(i, j) \in [K]^2$, the values $a_1, \ldots, a_K$ and $b_1, \ldots, b_K$ form a solution of the Grid Tiling instance.

Thus, the construction is a polynomial-time reduction from Grid Tiling. Since Grid Tiling is NP-hard, Single-Move Labeled Token Routing is NP-hard on trees of maximum degree three. Together with membership in NP, this proves NP-completeness.

For the parameterized statement, map the Grid Tiling instance to the pair (J_{GT}, μ) with $\mu = 3K^2 + 4K$; this bound is output on every input, whether or not the Grid Tiling instance is solvable, and it is bounded by a computable function of K. If the Grid Tiling instance is solvable, then J_{GT} admits a solution with solution edge multiplicity at most μ by Lemma 23. Conversely, the backward direction above converts an arbitrary solution of J_{GT}, so in particular one with solution edge multiplicity at most μ, into a solution of the Grid Tiling instance. Hence, (J_{GT}, μ) is a yes-instance of the parameterized problem if and only if the Grid Tiling instance is a yes-instance, so the construction is a parameterized reduction from Grid Tiling, parameterized by K. Since Grid Tiling is W[1]-hard when parameterized by K, the claimed W[1]-hardness follows. ◀

Prescribed movement order. It remains to prove the part of Theorem 4 concerning the variant in which the input additionally specifies an ordering of the tokens. Every pair of a compatible matching contains exactly one source, and every token initially occupies one source, so an ordering of the tokens induces a movement order for every compatible matching; a solution of the variant must use the induced movement order. The same reduction proves both hardness results for this variant.

Indeed, augment each constructed instance with the ordering of its tokens obtained by concatenating the following five sequences, in this order.

1. The sealing tokens; those associated with the same storage path appear in decreasing order of the distance between their storage targets and the vertex at which that path is attached to the rest of the tree.
2. The column tile tokens; those of the same value subtree appear in increasing order of the distance between their sources and the source of the sealing token of that value subtree.
3. The row tile tokens, ordered in the same way.

4. The cleanup tokens carrying labels of the form clean(i, j), in increasing order of the distance between their sources and the vertex at which the cleanup path is attached to the rest of the tree.
5. The remaining cleanup tokens, ordered in the same way.

Ties are broken by an arbitrary fixed ordering of the vertices.

This ordering is determined solely by the constructed instance and does not depend on the values of a solution of the Grid Tiling instance. The induced movement order is feasible for the route family of the solution constructed in the proof of Lemma 23 from any Grid Tiling solution. In both orders, all sealing tokens move before tile tokens, all column tile tokens move before row tile tokens, and all row tile tokens move before cleanup tokens, and the orders within value subtrees and paths are those used in that proof to ensure that the relevant routes are clear. Conversely, every solution of the variant is also a solution of the unrestricted instance, so the reverse direction applies unchanged. The ordering does not affect the forward solution, whose solution edge multiplicity remains at most $3K^2 + 4K$.

Therefore, even when a movement order is given as part of the input, Single-Move Labeled Token Routing is NP-complete on trees of maximum degree three and W[1]-hard when parameterized by the input bound μ on the solution edge multiplicity.

5.4 Candidate edge multiplicity on trees of unbounded degree

Our last hardness result shows that bounding the candidate edge multiplicity alone does not make the problem tractable: on trees of unbounded degree, Single-Move Labeled Token Routing remains NP-complete even when the candidate edge multiplicity is at most eight.

We reduce from 4-Bounded 3-SAT. An instance consists of a Boolean formula φ in conjunctive normal form in which every clause contains exactly three literals of three distinct variables, and every variable occurs at most four times. The question is whether φ admits a satisfying assignment. This problem is NP-complete [48]. We use the terminology for occurrences of Section 5.2: an occurrence of a literal of x or $\bar{x}$ in a clause C is an occurrence *of* the variable x and *in* the clause C, it is *positive* or *negative* according to its literal, and it is *true* or *false* under a truth assignment according to the value of its literal. Since the three literals of a clause have distinct variables, a variable has at most one occurrence in a clause and at most four occurrences in total, and a clause has exactly three occurrences.

To encode φ as a tree, we construct one *variable gadget* for each variable and one *clause gadget* for each clause. All gadgets are attached to a common ordinary vertex g, whose degree is therefore the number of variables plus the number of clauses. Every vertex other than g has constant degree, and the candidate edge multiplicity is at most eight, for two reasons that Lemma 26 verifies once the construction is in place: a gadget has constant size, because a variable has at most four occurrences and a clause has three, and, for each occurrence, only two compatible pairs have their source and their target in different gadgets.

Figure 7 shows a variable gadget and a clause gadget. We first describe the tokens of each gadget and the targets they may occupy; the vertices, edges, and labels of the gadgets are specified in the construction below. The variable gadget for a variable x determines the truth value of x. Its tokens are the *assignment token* a_x and the *complement token* $\bar{a}_x$, and its targets are the two *assignment targets* true_x and false_x. The labels will ensure that, in every compatible matching, a_x is matched with one assignment target and $\bar{a}_x$ with the other. Matching a_x with true_x represents setting x to `true`; matching it with false_x represents setting x to `false`. The clause gadget for a clause C determines which occurrence in C is designated to satisfy C. Its tokens are a *choice token* c_o for each occurrence o in C

and two *filler tokens*, and its targets are a *selection target* sel_o for each occurrence o in C and two *non-selection targets*. The labels will ensure that, in every compatible matching, exactly one choice token is matched with its selection target, the other two choice tokens are matched with the two non-selection targets, and the two filler tokens are matched with the two selection targets left free. The occurrence whose choice token is matched with its selection target is *selected*; the other two occurrences are *unselected*. We will show that a solution exists only if the selected occurrence of C is true.

Each occurrence o of a variable x in a clause C contributes two further tokens, a *literal token* ℓ_o and a *witness token* w_o, and these are the only tokens whose routes leave a gadget. Their routes are placed so that, as we will show, they impose precedence constraints between a_x and c_o. The token a_x must move before ℓ_o, and, if o is selected, ℓ_o must move before c_o. The token c_o must move before w_o, and, if o is false, w_o must move before a_x. If o is both selected and false, the four constraints form a cycle, so no feasible movement order exists. A compatible matching thus determines a truth assignment and one selected occurrence per clause, and we will show that it admits a feasible movement order exactly when every selected occurrence is true.

Construction. Tokens are named by the initials of their roles (a, c, ℓ, and w for *assignment*, *choice*, *literal*, and *witness*, respectively; the complement token $\bar{a}_x$ is named after the assignment token it complements). For each token z, we write $\mathrm{s}(z)$ for its source and, when the label of z is private, so that its target is determined, $\mathrm{t}(z)$ for its target. The other targets share their compatible sources with another target and are named individually. When occurrences are numbered $o_1, o_2, \ldots$, as in Figure 7, we write ℓ_i, w_i, c_i, and sel_i for ℓ_{o_i}, w_{o_i}, c_{o_i}, and sel_{o_i}.

Variable gadgets. For each variable x, we introduce an ordinary vertex g_x adjacent to g and, adjacent to g_x, the sources $\mathrm{s}(a_x)$ and $\mathrm{s}(\bar{a}_x)$ and the assignment targets true_x and false_x. For each occurrence o of x, we introduce the source $\mathrm{s}(\ell_o)$ of the literal token, adjacent to $\mathrm{s}(a_x)$, and the target $\mathrm{t}(w_o)$ of the witness token, adjacent to false_x if o is positive and to true_x if o is negative. The target of ℓ_o and the source of w_o lie in the clause gadget of the clause of o.

Clause gadgets. For each clause C, we introduce an ordinary vertex g_C adjacent to g. For each occurrence o in C, we introduce the source $\mathrm{s}(c_o)$ of the choice token, the source $\mathrm{s}(w_o)$ of the witness token, the selection target sel_o, and the target $\mathrm{t}(\ell_o)$ of the literal token, forming the two paths $g_C, \mathrm{s}(c_o), \mathrm{s}(w_o)$ and $g_C, \mathrm{sel}_o, \mathrm{t}(\ell_o)$. Finally, we introduce the sources of the two filler tokens of C and the two non-selection targets of C, all four adjacent to g_C.

For ease of reference, we list, for each label, the sources and targets that carry it and the compatibilities that it creates; in Figure 7, the sources and targets carrying one shared label are surrounded by one enclosure. No label is carried by any source or target other than those listed.

Assignment labels. The label $\mathrm{assign}(x)$, for each variable x, labels the sources $\mathrm{s}(a_x)$ and $\mathrm{s}(\bar{a}_x)$ and the targets true_x and false_x. Thus, a_x and $\bar{a}_x$ are each compatible with both assignment targets of x and with no other target, and each assignment target is compatible only with these two sources.

Literal labels. The label $\mathrm{literal}(o)$, for each occurrence o, is private: it labels only $\mathrm{s}(\ell_o)$ and $\mathrm{t}(\ell_o)$. It forces ℓ_o to move from the variable gadget of its variable to the clause gadget of its clause.

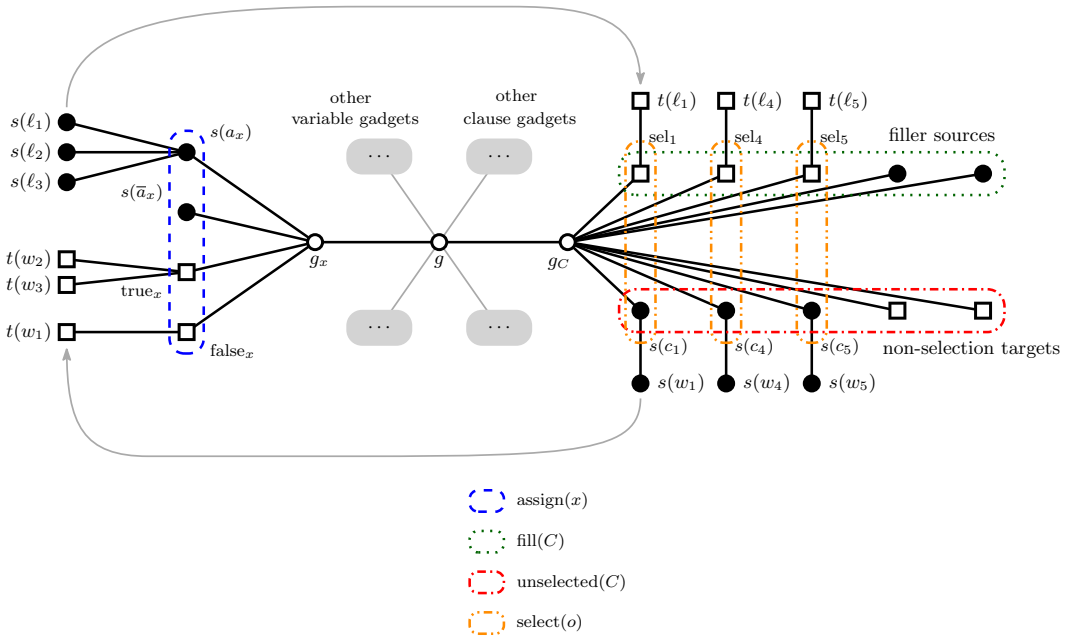


Figure 7 Part of the construction of Theorem 5: the variable gadget for a variable x with occurrences o_1, o_2, and o_3, of which o_1 is positive and lies in the clause C while o_2 and o_3 are negative and lie in two other clauses, and the clause gadget for C, whose occurrences are o_1, o_4, and o_5. A dashed enclosure surrounds the sources and targets that carry one shared label, and a vertex inside two enclosures carries both. The two arrows mark the two tokens of o_1 that move between the gadgets.

Witness labels. The label witness(o), for each occurrence o, is likewise private: it labels only $\mathrm{s}(w_o)$ and $\mathrm{t}(w_o)$, and forces w_o to move from the clause gadget to the variable gadget.

Selection labels. The label select(o), for each occurrence o, labels $\mathrm{s}(c_o)$ and sel_o.

Filling labels. The label fill(C), for each clause C, labels the sources of the two filler tokens of C and the selection target sel_o for each occurrence o in C. Thus, each filler token of C is compatible with the three selection targets of C, and sel_o is compatible with c_o and with the two filler tokens of C.

Non-selection labels. The label unselected(C), for each clause C, labels $\mathrm{s}(c_o)$ for each occurrence o in C and the two non-selection targets of C. Thus, each choice token of C is compatible with its selection target and with the two non-selection targets of C, and each non-selection target is compatible with the three choice tokens of C.

Let $J_{3\mathrm{SAT}} = (G, S, T, L)$ denote the constructed instance. For an occurrence o, we call $(\mathrm{s}(\ell_o), \mathrm{t}(\ell_o))$ the *literal pair* and $(\mathrm{s}(w_o), \mathrm{t}(w_o))$ the *witness pair* of o. Both pairs are compatible, and, since their labels are private, both belong to every compatible matching of $J_{3\mathrm{SAT}}$. We first establish the local properties of $J_{3\mathrm{SAT}}$, among them the bound on its candidate edge multiplicity.

▶ **Lemma 26.** *The instance $J_{3\mathrm{SAT}}$ can be computed in polynomial time, G is a tree, and $\mathrm{cm}(J_{3\mathrm{SAT}}) \leq 8$. Moreover, every source is compatible with at most three targets, every target is compatible with at most three sources, and every vertex other than g has degree at most eleven.*

Proof. Each variable, clause, and occurrence contributes a constant number of vertices, edges, and labels, so $|J_{3\mathrm{SAT}}|$ is linear in $|\varphi|$ and $J_{3\mathrm{SAT}}$ can be computed in polynomial time.

Every gadget is a tree, every gadget is joined to the rest of G by a single edge incident with g, and no edge joins two gadgets; hence G is a tree.

The compatibility bounds follow from the label list: a_x and $\bar{a}_x$ are compatible with two targets, ℓ_o and w_o with one, and every choice token and every filler token with three. Symmetrically, every assignment target is compatible with two sources, $\mathrm{t}(\ell_o)$ and $\mathrm{t}(w_o)$ with one, and every selection target and every non-selection target with three. For the degrees, g_C is adjacent to g, to the three choice sources, the three selection targets, the two filler sources, and the two non-selection targets of its gadget, and to nothing else, so it has degree 11; the vertices g_x, $\mathrm{s}(a_x)$, true_x, and false_x are adjacent to at most five vertices, since x has at most four occurrences; every remaining vertex other than g has degree at most 2.

We bound the candidate edge multiplicity edge by edge, counting every compatible pair whose path uses the edge, whether or not the pair belongs to a solution. Deleting an edge e of G leaves two components, and the path of a compatible pair uses e exactly when the pair has one endpoint in each of them; since g lies in one of the two components, the pairs whose path uses e are those with exactly one endpoint in the component of $G - e$ that does not contain g. By the label list, the only compatible pairs whose source and target lie in different gadgets are the literal pair and the witness pair of each occurrence; each remaining compatible pair joins two vertices of one gadget that are both adjacent to g_x or both adjacent to g_C. For $e = gg_x$, the component of $G - e$ not containing g is the variable gadget of x, which contains an endpoint of the literal pair and of the witness pair of each of the at most 4 occurrences of x and no endpoint of any other literal or witness pair, so gg_x lies on at most 8 paths; likewise, gg_C lies on 6 paths. For an edge $e = g_x v$ with v one of $\mathrm{s}(a_x)$, $\mathrm{s}(\bar{a}_x)$, true_x, and false_x, the component of $G - e$ not containing g consists of v and its leaves, so e lies on the paths of the 2 compatible pairs that contain v and a vertex of the gadget, and of at most 4 literal or witness pairs ending at leaves of v, hence on at most 6 paths. For an edge $e = g_C v$ with v another vertex of the clause gadget, the component of $G - e$ not containing g consists of v and its leaf, if it has one, so e lies on the paths of the at most 3 compatible pairs that contain v and a vertex of the gadget, and of at most 1 literal or witness pair, hence on at most 4 paths. Every remaining edge is incident with a leaf and lies on exactly 1 path. Therefore, $\mathrm{cm}(J_{3\mathrm{SAT}}) \leq 8$. ◀

We now turn to correctness. Since G is a tree, a compatible matching M determines its route family and hence the precedence digraph of Section 4, which we denote by $D(M)$. Its vertices are the pairs of M; to simplify notation, we name each pair by its token, so that an arc from a token to another means that the former must move before the latter under M. By Observation 10, M is the matching of a solution if and only if $D(M)$ is acyclic.

▶ **Lemma 27.** *Let M be a compatible matching of $J_{3\mathrm{SAT}}$.*

1. *For each variable x, M contains either $(\mathrm{s}(a_x), \mathrm{true}_x)$ and $(\mathrm{s}(\bar{a}_x), \mathrm{false}_x)$, or $(\mathrm{s}(a_x), \mathrm{false}_x)$ and $(\mathrm{s}(\bar{a}_x), \mathrm{true}_x)$.*
2. *For each clause C, there is exactly one occurrence o in C with $(\mathrm{s}(c_o), \mathrm{sel}_o) \in M$.*

Proof. Fix a variable x. By the assignment labels, $\mathrm{s}(a_x)$ and $\mathrm{s}(\bar{a}_x)$ are compatible only with true_x and false_x, and these two targets are compatible with no other source. Since M matches all four vertices, its restriction to these four vertices is one of the two stated possibilities.

Fix a clause C. By the non-selection labels, the two non-selection targets of C are compatible only with the three choice tokens of C, so M matches them with two distinct choice tokens. The remaining choice token c_o is compatible only with sel_o besides the two non-selection targets, so $(\mathrm{s}(c_o), \mathrm{sel}_o) \in M$. The two filler tokens of C are compatible only

with the selection targets of C, so M matches them with the two selection targets other than sel_o, and o is the only occurrence in C whose choice token is matched with its selection target. ◀

Let M be a compatible matching. By Lemma 27, every clause C has exactly one occurrence o with $(\mathrm{s}(c_o), \mathrm{sel}_o) \in M$; we call it the *selected* occurrence of C, and the other two occurrences in C *unselected*. By the same lemma, M matches $\mathrm{s}(a_x)$ with true_x or with false_x for every variable x, so M determines a truth assignment τ_M by setting $\tau_M(x) = \texttt{true}$ in the first case and $\tau_M(x) = \texttt{false}$ in the second.

For the reverse direction of the reduction, we will show that if M is the matching of a solution, then τ_M satisfies φ. Since every clause has a selected occurrence, it suffices to show that $D(M)$ has a directed cycle whenever some selected occurrence is false under τ_M: by Observation 10, the matching of a solution then has no false selected occurrence, so τ_M satisfies every clause. The next lemma gives the arcs of $D(M)$ that ℓ_o and w_o create, and shows that, when o is selected but false under τ_M, four of them form such a cycle. Two of the four exist in every compatible matching; the other two depend on the choices made in M, one on whether c_o is matched with sel_o and one on whether a_x is matched with the assignment target adjacent to $\mathrm{t}(w_o)$.

▶ **Lemma 28.** *Let M be a compatible matching of $J_{3\mathrm{SAT}}$, and let o be an occurrence of a variable x. Then $D(M)$ contains the arcs (a_x, ℓ_o) and (c_o, w_o); it contains the arc (ℓ_o, c_o) if o is selected, and the arc (w_o, a_x) if o is false under τ_M. Consequently, if o is selected and false under τ_M, then $D(M)$ contains a directed cycle, and M admits no feasible movement order.*

Proof. Let C be the clause of o. By construction, the route of ℓ_o contains $\mathrm{s}(a_x)$ and sel_o as internal vertices, and the route of w_o contains $\mathrm{s}(c_o)$ and the assignment target adjacent to $\mathrm{t}(w_o)$ as internal vertices. By the definition of $D(M)$, the sources $\mathrm{s}(a_x)$ and $\mathrm{s}(c_o)$ on these routes give the arcs (a_x, ℓ_o) and (c_o, w_o); which arcs the two remaining vertices give depends on the tokens that M matches with them.

If o is selected, then sel_o is the target of c_o and lies on the route of ℓ_o, so $D(M)$ contains the arc (ℓ_o, c_o). The assignment target adjacent to $\mathrm{t}(w_o)$ is false_x if o is positive and true_x if o is negative, so by Lemma 27 and the definition of τ_M it is the target of a_x exactly when o is false under τ_M; since it lies on the route of w_o, $D(M)$ contains the arc (w_o, a_x) in that case.

If o is selected and false under τ_M, the four arcs form the directed cycle $a_x, \ell_o, c_o, w_o, a_x$ in $D(M)$, so by Observation 10 no feasible movement order exists for M. ◀

We now have all the statements needed to prove this subsection's main result.

▶ **Theorem 5.** *Single-Move Labeled Token Routing is NP-complete on trees with candidate edge multiplicity at most eight, even when every source is compatible with at most three targets and every target with at most three sources.*

Proof. By Lemma 26, $J_{3\mathrm{SAT}}$ can be computed in polynomial time, G is a tree, $\mathrm{cm}(J_{3\mathrm{SAT}}) \leq 8$, every source is compatible with at most three targets, and every target is compatible with at most three sources. We show that φ is satisfiable if and only if $J_{3\mathrm{SAT}}$ is a yes-instance.

($\Leftarrow$) Suppose that $J_{3\mathrm{SAT}}$ is a yes-instance, and let M be the compatible matching of a solution. We show that τ_M, the truth assignment that sets x to $\texttt{true}$ if M matches $\mathrm{s}(a_x)$ with true_x and to $\texttt{false}$ if it matches $\mathrm{s}(a_x)$ with false_x, satisfies φ. Fix a clause C, and let o be its selected occurrence, which exists by Lemma 27. If o were false under τ_M, then

by Lemma 28 the matching M would admit no feasible movement order, contradicting the choice of M. Hence, o is true under τ_M, so τ_M satisfies C. Since this holds for every clause, τ_M satisfies φ.

($\Rightarrow$) Suppose that φ has a satisfying assignment τ. Define a compatible matching M as follows. For each variable x, include $(\mathrm{s}(a_x), \mathrm{true}_x)$ and $(\mathrm{s}(\bar{a}_x), \mathrm{false}_x)$ if $\tau(x) = \texttt{true}$, and $(\mathrm{s}(a_x), \mathrm{false}_x)$ and $(\mathrm{s}(\bar{a}_x), \mathrm{true}_x)$ otherwise. For each occurrence o, include the literal pair and the witness pair of o. For each clause C, fix an occurrence o in C that is true under τ; include $(\mathrm{s}(c_o), \mathrm{sel}_o)$, match the other two choice tokens of C with the two non-selection targets of C, and match the two filler tokens of C with the two remaining selection targets of C. Each source and each target occurs in exactly one pair, and every pair is compatible, so M is a compatible matching with $\tau_M = \tau$ in which the occurrence fixed for each clause is selected. In particular, every selected occurrence is true under τ. We now construct a feasible movement order. We move the tokens in seven stages, moving all tokens of a stage, in any order, before those of the next; the vertices g, g_x, and g_C are ordinary and hence never occupied.

1. Move every choice token of an unselected occurrence to its non-selection target, along a route whose only internal vertex is g_C. The route is clear.
2. Move every witness token w_o of an unselected occurrence o that is false under τ. Its route passes through $\mathrm{s}(c_o)$, vacated in stage 1, and through the assignment target adjacent to $\mathrm{t}(w_o)$, which is still unoccupied because no token of a variable gadget has moved; its target $\mathrm{t}(w_o)$ is unoccupied. The route is clear.
3. Move every assignment token a_x to the assignment target matched with it, along a route whose only internal vertex is g_x. The witness targets occupied in stage 2 are leaves adjacent to that assignment target and do not lie on this route, and the assignment target itself is unoccupied.
4. Move every literal token ℓ_o. Its route passes through $\mathrm{s}(a_x)$, where x is the variable of o, vacated in stage 3, and through sel_o and $\mathrm{t}(\ell_o)$, which are still unoccupied because no choice token or filler token has reached a selection target. The route is clear.
5. For each clause C with selected occurrence o, move c_o along $\mathrm{s}(c_o), g_C, \mathrm{sel}_o$; the route ends at sel_o and does not contain the occupied vertex $\mathrm{t}(\ell_o)$. Then move the two filler tokens of C to the two remaining selection targets of C, along routes whose only internal vertex is g_C.
6. Move every witness token w_o that has not yet moved. The occurrence o is true under τ, so the assignment target adjacent to $\mathrm{t}(w_o)$ is matched with $\bar{a}_x$, where x is the variable of o, and is still unoccupied; $\mathrm{s}(c_o)$ has been vacated, in stage 1 if o is unselected and in stage 5 if it is selected. The route is clear.
7. Move every complement token $\bar{a}_x$ to the assignment target not taken by a_x, which is unoccupied, along a route whose only internal vertex is g_x; the witness targets occupied in stage 6 are leaves adjacent to that assignment target and do not lie on the route.

Every token moves exactly once along the unique path of its pair in M. Hence M, together with this movement order, is a solution, and $J_{3\mathrm{SAT}}$ is a yes-instance.

Thus, the construction is a polynomial-time reduction from 4-Bounded 3-SAT, so Single-Move Labeled Token Routing is NP-hard under the stated restrictions. Together with membership in NP, this proves NP-completeness. ◀

The high degree of the construction is concentrated at g, and Lemma 11 cannot remove it without losing the bound on the candidate edge multiplicity. The lemma preserves feasibility,

but replacing g by a tree F_g of maximum degree three routes the literal and witness pairs of all occurrences through the edges of F_g, and an edge of F_g then lies on the path of every such pair whose two gadgets it separates, a number that is not bounded by a constant for an arbitrary formula. Indeed, by Theorem 8, bounded degree together with bounded candidate edge multiplicity yields tractability, so any hardness result with bounded candidate edge multiplicity requires unbounded maximum degree, unless $\mathsf{P} = \mathsf{NP}$.

6 Positive results beyond the all-matching case

The hardness results of the previous section show that restricting the underlying graph to a tree of maximum degree three is not, by itself, sufficient to obtain tractability. We first show that Single-Move Labeled Token Routing can be solved efficiently on paths and when the number of tokens is bounded. We then return to trees under a bound on the candidate edge multiplicity together with the maximum degree; informally, bounding the candidate edge multiplicity limits the number of possible matching choices between the sources and the targets across any edge of the tree. Finally, we show that the same algorithm yields fixed-parameter tractability when the problem is parameterized by the candidate vertex multiplicity alone.

6.1 Paths

If an instance on a path admits a solution, then the matching used by that solution is forced. Thus, it suffices to determine this matching, test the compatibility of its pairs, and compute a feasible movement order.

▶ **Theorem 6.** *Single-Move Labeled Token Routing on a path can be solved in time* $\mathcal{O}(|I|)$.

Proof. Let P be the underlying path, designate one of its endpoints as the left endpoint, and number the sources from left to right as $s_1, \ldots, s_k$ and the targets from left to right as $t_1, \ldots, t_k$. The algorithm accepts if and only if s_i and t_i are compatible for every i.

We first show that a solution, if one exists, uses the matching $\{(s_i, t_i) : i \in [k]\}$. Consider the first move of an arbitrary solution and suppose it takes the token at s_i to t_j. If $s_i = t_j$, the route of this move is the one-vertex path at that vertex, which stays occupied throughout, so no other route uses it, and each remaining source is matched with a target on its own side of this vertex. Since $i - 1$ sources and $j - 1$ targets lie to its left, $i = j$, and the two sides are handled by induction. Assume henceforth that $s_i \neq t_j$. Once t_j is occupied, no later route may use it, so every remaining source on one side of t_j must be routed to a target on the same side. There are exactly $j - 1$ targets to the left of t_j. If s_i lies to the left of t_j, then exactly j sources lie to the left of t_j before the move, and since the route from s_i to t_j is clear, s_i is the last of them, so $i = j$. If s_i lies to the right of t_j, then exactly $j - 1$ sources lie to the left of t_j and the clear route forces s_i to be the first source to the right of t_j, so again $i = j$. The move therefore matches s_i to t_i, and it leaves all pairs of smaller index to the left of t_i and all pairs of larger index to the right of it. Applying the argument independently to the two sides and inducting yields the claim; consequently, if s_i and t_i are incompatible for some i, no solution exists.

Assume now that s_i and t_i are compatible for every i. We construct a feasible movement order for the matching $\{(s_i, t_i) : i \in [k]\}$. First move the tokens whose target lies to the right of their source, in decreasing order of their sources; then move the tokens whose target lies to the left of their source, in increasing order of their sources. We check that each route

is clear when its token moves, writing $u < v$ when the vertex u lies to the left of the vertex v. Let $s_i < t_i$. A source s_j with $s_i < s_j \le t_i$ has $j > i$, so $t_j > t_i \ge s_j$ and the token at s_j moved before the token at s_i; a target t_j with $s_i < t_j < t_i$ has $j < i$, so $s_j < s_i < t_j$ and the token at s_j has not moved yet. Now let $t_i < s_i$. A source s_j with $t_i \le s_j < s_i$ has $j < i$, so $t_j < t_i \le s_j$ and the token at s_j moved before the token at s_i; a target t_j with $t_i < t_j < s_i$ has $j > i$, so $t_j < s_i < s_j$ and the token at s_j has not moved yet. In both cases the route from s_i to t_i contains no occupied vertex other than s_i.

It remains to bound the running time. The sources and the targets are found by a single traversal of P in time $\mathcal{O}(n)$, and the movement order is produced in time $\mathcal{O}(n)$ by listing the tokens whose source precedes their target in decreasing order of i, followed by the remaining tokens in increasing order of i. What remains is to decide, for every i, whether the label set $L_S(s_i)$ of the source s_i and the label set $L_T(t_i)$ of the target t_i intersect. Assume, as the encoding allows, that the labels are integers in $[|\Lambda|] = \{1, \dots, |\Lambda|\}$, and initialize a Boolean array indexed by $[|\Lambda|]$ with every entry false, in time $\mathcal{O}(|I|)$. For each i in turn, set the entry of every label of $L_S(s_i)$ to true, scan $L_T(t_i)$ for a label whose entry is true, and reset the entries of $L_S(s_i)$ to false. Each label of each set is handled a constant number of times, so all k tests take time $\sum_{i=1}^{k} \big(|L_S(s_i)| + |L_T(t_i)|\big) = \mathcal{O}(|I|)$, and the algorithm runs in time $\mathcal{O}(|I|)$. ◀

6.2 A bounded number of tokens

By Theorem 4, the problem remains NP-complete even when a movement order is given as part of the input. When the number of tokens is small, however, both the matching and the movement order can be enumerated, and once both are chosen, finding the routes reduces to a sequence of reachability tests.

▶ **Theorem 7.** *Single-Move Labeled Token Routing is fixed-parameter tractable when parameterized by the number k of tokens.*

Proof. Let $I = (G, S, T, L)$ be an instance with $|S| = |T| = k$. Fix a compatible matching M, write $M(s)$ for the target matched with a source s, and fix an ordering $(s_1, \dots, s_k)$ of S. Suppose that the tokens initially placed at $s_1, \dots, s_{i-1}$ have been moved to $M(s_1), \dots, M(s_{i-1})$ and that no other token has moved. Then the occupied vertices other than s_i are exactly the vertices $M(s_j)$ with $j < i$ and s_j with $j > i$, whatever routes were used. Hence, there is a clear s_i-$M(s_i)$ route at this point if and only if $M(s_i)$ is reachable from s_i in the graph G_i obtained from G by deleting these vertices, and I is a yes-instance if and only if, for some compatible matching M and some ordering of S, this holds for every $i \in [k]$; the routes are then any s_i-$M(s_i)$ paths in the graphs G_i.

The algorithm enumerates every bijection M from S to T, discards it unless $L_S(s) \cap L_T(M(s)) \neq \emptyset$ for every $s \in S$, and, for each remaining M and each ordering $(s_1, \dots, s_k)$ of S, tests by breadth-first search whether $M(s_i)$ is reachable from s_i in G_i for every $i \in [k]$. It accepts if and only if all k tests succeed for some M and some ordering, which is correct by the previous paragraph. There are $k!$ bijections and $k!$ orderings, the compatibility check takes time polynomial in $|I|$, and each of the k breadth-first searches takes time $\mathcal{O}(n + m)$, so the running time is $(k!)^2 \cdot |I|^{\mathcal{O}(1)}$, which is of the form $f(k) \cdot |I|^{\mathcal{O}(1)}$. ◀

6.3 Bounded candidate edge multiplicity on bounded-degree trees

The candidate edge multiplicity bounds, for each edge of the tree, the number of compatible source-target pairs whose unique paths use that edge. After rooting the tree, this bounds

the number of pairs of a matching whose paths leave any rooted subtree. Bounded degree then bounds the number of child subtrees that are combined at each vertex. Together, these two bounds yield, via dynamic programming, a fixed-parameter algorithm for instances on trees in which both the candidate edge multiplicity and the maximum degree of the tree are bounded. We first present the notation and describe the algorithm. We then prove its correctness (Lemma 31) and bound its running time (Lemma 33) in separate statements, since Section 6.4 reuses the algorithm and its correctness proof but bounds its running time by a different parameter.

Rooted tree, candidate pairs, and partial matchings. Let $I = (G, S, T, L)$ be a tree instance, and let Δ be the maximum degree of G. For a compatible pair $q = (s, t)$ we write P_q for the unique s-t path in G, which is the route of q in every solution whose matching contains q, and we say that q *crosses* an edge if P_q uses that edge. Root G at an arbitrary vertex r. For each vertex v, let G_v denote the subtree rooted at v. For each vertex v other than r, let e_v be the edge joining v to its parent, called the *boundary edge* of G_v. We call v the interface vertex of G_v: every path of a pair crossing e_v meets G_v at v before crossing e_v. Let $\mathrm{Cand}(v)$ be the set of compatible pairs that cross e_v, called the *candidate pairs* of v. Thus, $\mathrm{Cand}(v)$ consists of the compatible pairs with one endpoint in G_v and the other outside G_v, and $|\mathrm{Cand}(v)| = \mathrm{cm}_I(e_v) \le \mathrm{cm}(I)$ by the definition of candidate edge multiplicity. For the root we set $\mathrm{Cand}(r) = \emptyset$. A set of compatible pairs is a *partial matching* if every source occurs in at most one pair and every target occurs in at most one pair.

Description of the entries in the dynamic programming table. A subtree interacts with the rest of the tree only through compatible pairs whose paths cross its boundary edge. The interface vertex is the point at which all such paths leave the subtree, and the table stores the boundary pairs together with the precedence relation forced among them. For each vertex v, the table contains one entry $\mathrm{DP}[v, X, \prec]$ for each partial matching $X \subseteq \mathrm{Cand}(v)$ and each strict partial order $\prec$ on X, that is, each transitive and irreflexive relation on X. Every entry is either true or false. The set X stands for the pairs of a partial matching that cross e_v, and $p \prec q$ stands for the requirement, imposed by the pairs of that matching inside G_v, that the move of p precede the move of q.

We say that a partial matching M *realizes* the state $(X, \prec)$ at v if it satisfies the following conditions.

1. **Coverage.** Every source in $S \cap V(G_v)$ appears as the source of exactly one pair of M, and every target in $T \cap V(G_v)$ appears as the target of exactly one pair of M.
2. **Locality.** Every pair of M has at least one endpoint in $V(G_v)$.
3. **Boundary.** $M \cap \mathrm{Cand}(v) = X$. For $v \neq r$, this means that the pairs of M that cross e_v are precisely the pairs of X; for $v = r$, it means that $X = \emptyset$.
4. **Precedence.** Let $D_v(M)$ be the digraph with vertex set M in which, for two distinct pairs $p = (s_p, t_p)$ and $q = (s_q, t_q)$ of M, there is an arc (p, q), meaning that the move of p must precede the move of q, if $s_p \in V(G_v) \cap V(P_q)$ or $t_q \in V(G_v) \cap V(P_p)$; we say that such an arc is *witnessed in* G_v. The digraph $D_v(M)$ is acyclic, and, for $p, q \in X$, we have $p \prec q$ if and only if $D_v(M)$ contains a directed path from p to q. Informally, $p \prec q$ precisely when the matching inside G_v forces p to be moved before q.

The intended invariant is that $\mathrm{DP}[v, X, \prec]$ is true if and only if some partial matching realizes $(X, \prec)$ at v. We write $\emptyset$ both for the empty set and for the empty order.

Initialization. Let v be a vertex with no children. We mark the following entries as true; every state not listed below is set to false.

1. If $v \notin S \cup T$, then $\mathrm{DP}[v, \emptyset, \emptyset]$ is true.
2. If $v \in S \setminus T$, then, for each pair $(v,t) \in \mathrm{Cand}(v)$, the entry $\mathrm{DP}[v, \{(v,t)\}, \emptyset]$ is true.
3. If $v \in T \setminus S$, then, for each pair $(s,v) \in \mathrm{Cand}(v)$, the entry $\mathrm{DP}[v, \{(s,v)\}, \emptyset]$ is true.
4. Suppose that $v \in S \cap T$. If (v,v) is compatible, then $\mathrm{DP}[v, \emptyset, \emptyset]$ is true. In addition, for every two pairs $(v,t), (s,v) \in \mathrm{Cand}(v)$ that together form a partial matching, the entry $\mathrm{DP}[v, X, \prec]$ with $X = \{(v,t), (s,v)\}$ and $(v,t) \prec (s,v)$ is true, the token at v having to leave before the token assigned to v arrives.

Combining child states. Let v be a vertex with at least one child. For each child u of v, choose a true entry $\mathrm{DP}[u, X_u, \prec_u]$; choose also a partial matching $X \subseteq \mathrm{Cand}(v)$. If $v \in S \cap T$, we additionally choose either the compatible self-pair (v,v) or no self-pair. Let $Y = X \cup \bigcup_u X_u$, where u ranges over the children of v. These are the pairs whose routes use an edge incident with v. We check the following three conditions on Y. If all three hold, we compute an order $\prec$ on X and mark $\mathrm{DP}[v, X, \prec]$ as true.

1. **Endpoint conditions.** The set Y is a partial matching. Moreover, for each child u of v, the set X_u consists precisely of the pairs in Y having exactly one endpoint in $V(G_u)$. If v is not the root, then X consists precisely of the pairs in Y having exactly one endpoint in $V(G_v)$.
2. **Requirements at v.** If $v \notin S \cup T$, then no pair in Y has source or target v. If $v \in S \setminus T$, then exactly one pair in Y has source v, and no pair in Y has target v. Symmetrically, if $v \in T \setminus S$, then exactly one pair in Y has target v, and no pair in Y has source v. If $v \in S \cap T$, then there are two possibilities. If the self-pair (v,v) is chosen, then no pair in Y has source or target v, and no pair in Y uses v as an internal vertex of its path. Otherwise, exactly one pair in Y has source v, and exactly one pair in Y has target v.
3. **Acyclicity.** Let D be the digraph with vertex set Y whose arcs are the following. For each child u of v and all $p, q \in X_u$ with $p \prec_u q$, the pair (p,q) is an arc. For all distinct $p = (s_p, t_p)$ and $q = (s_q, t_q)$ in Y, the pair (p,q) is an arc whenever $s_p = v$ and $v \in V(P_q)$, or $t_q = v$ and $v \in V(P_p)$; we say that such an arc is *witnessed at* v. The digraph D is acyclic.

We define $\prec$ on X by setting $p \prec q$ if and only if D contains a directed path from p to q; since D is acyclic, $\prec$ is a strict partial order. Every entry $\mathrm{DP}[v, X, \prec]$ not marked true by any choice is false. Note that the entry $\mathrm{DP}[v, X, \prec]$ keeps only X and not all of Y. By the endpoint conditions, each pair of $Y \setminus X$ has both endpoints in G_v, so no later combination can interact with that pair except through the precedence relation that it induces on the boundary pairs. The computation above v therefore needs only X and $\prec$.

Figure 8 illustrates one combination step, at a vertex v with children u and w. The pairs p_1 and p_4 have an endpoint outside G_v and therefore cross e_v, the pair p_3 joins the two child subtrees, the pair p_2 has source v, and the pair p_5 lies inside G_u. In G_u, the route of p_1 contains $\mathrm{s}(p_5)$ and the route of p_3 contains $\mathrm{t}(p_5)$, so the arcs witnessed in G_u are (p_5, p_1) and (p_3, p_5). The pair p_5 does not cross e_u, so the entry chosen at u has $X_u = \{p_1, p_3\}$ and $p_3 \prec_u p_1$. In G_w, the route of p_3 contains $\mathrm{s}(p_4)$, so the entry chosen at w has $p_4 \prec_w p_3$. Since v is the source of p_2 and lies on the routes of p_1, p_3, and p_4, the arcs (p_2, p_1), (p_2, p_3), and (p_2, p_4) are witnessed at v. The digraph D consists of these three arcs together with (p_3, p_1) and (p_4, p_3), and it is acyclic. Only p_1 and p_4 cross e_v, so $X = \{p_1, p_4\}$, and the directed path p_4, p_3, p_1 of D gives $p_4 \prec p_1$, an order contained in neither child entry.

Order of computation and acceptance. The algorithm first computes the path P_q of each compatible pair q and the set $\mathrm{Cand}(v)$ of each vertex v, by enumerating the compatible

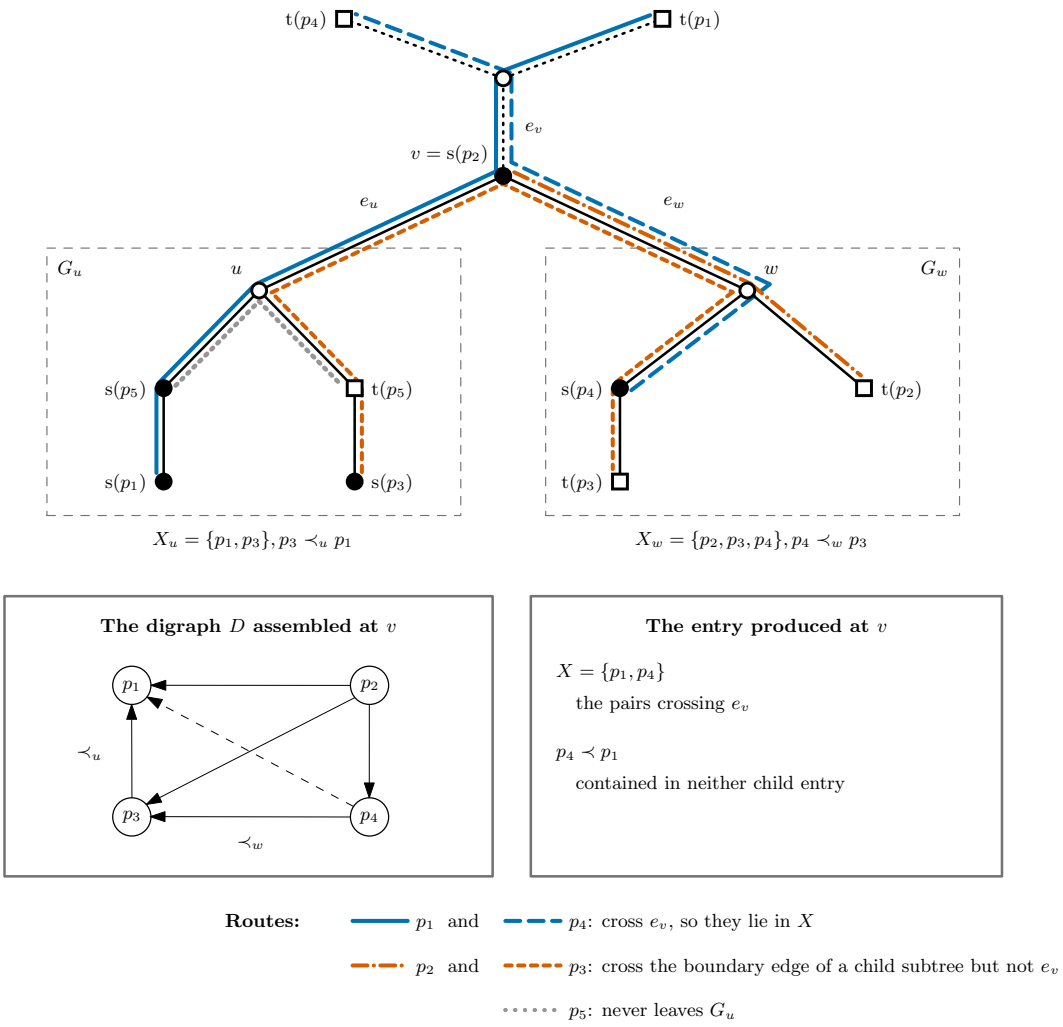


Figure 8 One combination step at a vertex v with children u and w, described in the text. The route of each pair of the matching is drawn beside the tree edges it uses; as the legend shows, its color and dash pattern indicate whether the pair crosses e_v, stops at a child boundary, or stays inside G_u. Below the tree, the entries chosen at u and w are listed under G_u and G_w, the left panel shows the digraph D, and the right panel shows the entry produced at v; the dashed arrow is the order $p_4 \prec p_1$.

pairs and, for each of them, the edges of its path; this takes time polynomial in $|I|$. It then processes the vertices in a bottom-up order, and computes the entries $\mathrm{DP}[v, X, \prec]$ of each vertex v by the initialization if v has no children and by combining child states otherwise. Finally, it accepts if and only if $\mathrm{DP}[r, \emptyset, \emptyset]$ is true; since $\mathrm{Cand}(r) = \emptyset$, this is the only entry of the form $\mathrm{DP}[r, X, \prec]$.

Correctness. We first show that the entries computed by the algorithm have the intended meaning.

▶ **Lemma 29.** *For each vertex v, after the algorithm has computed the entries $\mathrm{DP}[v, X, \prec]$, such an entry is true if and only if there exists a partial matching that realizes $(X, \prec)$ at v.*

Proof. We prove the statement by induction on the height of G_v.

Base case. Suppose that v has no children.

1. **Case** $v \notin S \cup T$**.** The empty matching is the only partial matching satisfying the coverage and locality conditions, and it realizes $(\emptyset, \emptyset)$.
2. **Case** $v \in S \setminus T$**.** The coverage condition requires exactly one pair with source v. Since every pair of the matching must have an endpoint in G_v, this pair crosses the boundary edge e_v and therefore belongs to $\mathrm{Cand}(v)$. Hence, the realizable states are precisely the states $(\{(v,t)\}, \emptyset)$ created in the second initialization case.
3. **Case** $v \in T \setminus S$**.** This case is symmetric to the case $v \in S \setminus T$.
4. **Case** $v \in S \cap T$**.** A realizing matching either contains the self-pair (v,v), which realizes $(\emptyset, \emptyset)$, or contains two pairs crossing e_v, one with source v and one with target v. In the latter case, the pair with source v is forced to precede the pair with target v. Thus, the realizable states are exactly the entries created in the fourth initialization case.

This proves the statement for vertices with no children.

Inductive step. Let v be a vertex with at least one child, and assume that the statement holds for every child of v.

Entries produced by the algorithm. Consider a choice of child entries and a set X that passes all conditions of the algorithm. By the inductive hypothesis, for each child u of v, there is a partial matching M_u that realizes $(X_u, \prec_u)$ at u.

Construct a set M by taking the union of the matchings M_u over all children u of v, treating equal pairs as a single pair, adding every pair in X that is not already present, and adding (v,v) when the self-pair is chosen.

The endpoint conditions ensure that every pair appears in the appropriate parts of this union. A pair with endpoints in two distinct child subtrees occurs in the matchings of the corresponding two children. A pair with one endpoint in a child subtree and one endpoint at v occurs only in the matching of that child. A pair with one endpoint in a child subtree and the other outside G_v occurs in the matching of that child and in X. Finally, a pair with one endpoint at v and the other outside G_v occurs only in X.

Since every M_u is a partial matching, since the child subtrees are disjoint, and since Y is a partial matching, the set M is a partial matching. Every source and target in a child subtree is covered by the corresponding child matching. The requirements at v ensure that the source and target roles of v, when present, are also covered exactly once. Thus, M satisfies the coverage condition.

Every pair inherited from M_u has an endpoint in G_u, every pair in X has an endpoint in G_v, and the only possible additional pair is (v,v). Hence, M satisfies the locality condition. The endpoint conditions also imply that $M \cap \mathrm{Cand}(v) = X$, which is the boundary condition.

It remains to verify the precedence condition.

▷ Claim 30. The digraph $D_v(M)$ is acyclic if and only if D is acyclic. Moreover, when both are acyclic, for all $p, q \in X$, the digraph $D_v(M)$ contains a directed path from p to q if and only if D does.

Proof. Every arc of $D_v(M)$ is witnessed either at v itself or in one child subtree. The arcs of $D_v(M)$ witnessed at v are precisely the arcs of D witnessed at v.

Now fix a child u of v. Any pair whose two endpoints belong to G_u has its entire path inside G_u. Hence, such a pair cannot impose a precedence constraint outside G_u. The only effect of the matching inside G_u on the rest of the tree is therefore the order that it forces among the pairs that cross the boundary edge of G_u, namely the pairs in X_u. Since M_u realizes $(X_u, \prec_u)$, this order is exactly $\prec_u$.

Thus, after replacing the part of the matching inside each child subtree by the corresponding order $\prec_u$, the digraph D contains all precedence requirements witnessed in G_v that can affect pairs outside the child subtrees. Any directed path that enters a child subtree through a boundary pair and leaves it again can be replaced by the corresponding relation $\prec_u$; a path that has both endpoints inside the child subtree is handled entirely by the child state. Consequently, $D_v(M)$ is acyclic if and only if D is acyclic. Moreover, for $p, q \in X$, the digraph $D_v(M)$ contains a directed path from p to q if and only if D does. ◀

By Claim 30, the algorithm rejects the current choice precisely when M violates the precedence condition. Otherwise, the order $\prec$ defined by the algorithm is exactly the order required in the definition of a state realized by M. Hence, every entry produced by the algorithm is realized by some partial matching.

Every realizable state is produced. Conversely, suppose that a partial matching M realizes $(X, \prec)$ at v. For each child u of v, let M_u consist of the pairs of M having at least one endpoint in G_u. Let $X_u = M_u \cap \mathrm{Cand}(u)$ be the set of pairs of M_u that cross the boundary edge of G_u, and, for $p, q \in X_u$, let $p \prec_u q$ if and only if $D_u(M_u)$ contains a directed path from p to q.

The matching M_u satisfies the coverage and the locality conditions for G_u, its pairs crossing the boundary edge of G_u are exactly the pairs in X_u, and $D_u(M_u)$ is acyclic, since it is a subdigraph of $D_v(M)$. Thus, M_u realizes $(X_u, \prec_u)$ at u. By the inductive hypothesis, $\mathrm{DP}[u, X_u, \prec_u]$ is true for each child u of v.

Choose these child entries and the set X in the algorithm. Since M is a partial matching, the set Y, the union of X and of the sets X_u, is also a partial matching. By the definition of the sets X_u and X, the endpoint conditions hold.

The requirements at v follow from the coverage condition. If $(v, v) \in M$, then no other pair of M has source or target v, because M is a partial matching. Moreover, no pair of M uses v as an internal vertex of its path: otherwise, the self-pair and that pair would form a precedence cycle witnessed at v. Hence, the self-pair case is valid. If $(v, v) \notin M$, then the coverage condition gives exactly the required number of pairs with source v and of pairs with target v.

Finally, Claim 30 shows that the digraph D constructed by the algorithm is acyclic and that, for $p, q \in X$, it contains a directed path from p to q if and only if $p \prec q$. Therefore, the algorithm sets $\mathrm{DP}[v, X, \prec]$ to true. This proves the inductive step. ◀

Applying Lemma 29 at the root yields the correctness of the algorithm.

▶ **Lemma 31.** *The algorithm accepts if and only if I is a yes-instance.*

Proof. The algorithm accepts if and only if $\mathrm{DP}[r, \emptyset, \emptyset]$ is true. By Lemma 29, this holds if and only if there exists a partial matching M that realizes $(\emptyset, \emptyset)$ at r. As $G_r = G$, the coverage condition implies that M is a compatible matching of all sources and targets. Moreover, the digraph $D_r(M)$ is the precedence digraph of M and of the route family formed by the unique paths P_q, and the precedence condition states that this digraph is acyclic. By Observation 10, such a compatible matching exists if and only if I admits a solution. Therefore, the algorithm accepts if and only if I is a yes-instance. ◀

Running time. We bound the running time in two steps. Lemma 32 bounds the number of entries $\mathrm{DP}[v, X, \prec]$ of a vertex v and the time needed to compute them in terms of the numbers of candidate pairs of v and of its children, without reference to the parameters, and Lemma 33 bounds these numbers by the candidate edge multiplicity and the maximum degree.

▶ **Lemma 32.** *Let v be a vertex of G, and let u range over the children of v. There are at most $2^{|\mathrm{Cand}(v)|^2+|\mathrm{Cand}(v)|}$ entries $\mathrm{DP}[v, X, \prec]$, and the algorithm computes them in time $2^{|\mathrm{Cand}(v)|+\sum_u(|\mathrm{Cand}(u)|^2+|\mathrm{Cand}(u)|)} \cdot |I|^{\mathcal{O}(1)}$.*

Proof. An entry $\mathrm{DP}[v, X, \prec]$ is indexed by a partial matching $X \subseteq \mathrm{Cand}(v)$ and a strict partial order $\prec$ on X. There are at most $2^{|\mathrm{Cand}(v)|}$ subsets X of $\mathrm{Cand}(v)$ and, for each of them, at most $2^{|X|^2} \leq 2^{|\mathrm{Cand}(v)|^2}$ relations on X, since a relation on X is a subset of $X \times X$. This counts all relations and therefore upper-bounds the number of strict partial orders stored by the algorithm. This proves the first statement.

If v has no children, the initialization examines every pair of $\mathrm{Cand}(v)$ and, when $v \in S \cap T$, every two pairs of $\mathrm{Cand}(v)$, which takes time polynomial in $|I|$. Otherwise, the algorithm examines every choice of one true entry $\mathrm{DP}[u, X_u, \prec_u]$ for each child u of v, of a partial matching $X \subseteq \mathrm{Cand}(v)$, and, when $v \in S \cap T$, of whether the self-pair (v, v) is chosen. By the first statement applied to each child u, the child entries can be chosen in at most $\prod_u 2^{|\mathrm{Cand}(u)|^2+|\mathrm{Cand}(u)|} = 2^{\sum_u(|\mathrm{Cand}(u)|^2+|\mathrm{Cand}(u)|)}$ ways, the set X in at most $2^{|\mathrm{Cand}(v)|}$ ways, and the self-pair in at most two ways. Every choice is processed in time polynomial in $|I|$. Indeed, v has fewer than n children, and every entry consists of at most $|S| \cdot |T|$ compatible pairs and a relation on them, so the chosen entries are read and the set Y is formed in time polynomial in $|I|$. The endpoint conditions and the requirements at v are then checked, and the digraph D on Y, its acyclicity, and the order $\prec$ on X are computed, in time polynomial in $|I|$, since $|Y| \leq |S| \cdot |T|$ and the paths of the compatible pairs have been computed beforehand. Multiplying the three counts by the polynomial time per choice yields the second statement. ◀

▶ **Lemma 33.** *The algorithm runs in time $2^{\mathcal{O}(\Delta \cdot \mathrm{cm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$.*

Proof. The paths of the compatible pairs and the sets $\mathrm{Cand}(v)$ are computed in time polynomial in $|I|$. Fix a vertex v of G, and let u range over the children of v. By the definition of candidate pairs, $|\mathrm{Cand}(w)| \leq \mathrm{cm}(I)$ for every vertex w of G, and v has at most Δ children, so $|\mathrm{Cand}(v)| + \sum_u(|\mathrm{Cand}(u)|^2 + |\mathrm{Cand}(u)|) \leq \mathrm{cm}(I) + \Delta\big(\mathrm{cm}(I)^2 + \mathrm{cm}(I)\big)$, which is $\mathcal{O}(\Delta \cdot \mathrm{cm}(I)^2)$. Hence, by Lemma 32, the entries $\mathrm{DP}[v, X, \prec]$ are computed in time $2^{\mathcal{O}(\Delta \cdot \mathrm{cm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$. Summing over the n vertices of G yields the claimed bound. ◀

▶ **Theorem 8.** *Single-Move Labeled Token Routing on a tree instance $I = (G, S, T, L)$ can be solved in time $2^{\mathcal{O}(\Delta \cdot \mathrm{cm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$, where Δ is the maximum degree of G and $\mathrm{cm}(I)$ is the candidate edge multiplicity of I. Consequently, the problem is fixed-parameter tractable when parameterized by $\Delta + \mathrm{cm}(I)$.*

Proof. By Lemma 31, the algorithm accepts if and only if I is a yes-instance. By Lemma 33, it runs in time $2^{\mathcal{O}(\Delta \cdot \mathrm{cm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$. Since $\Delta \cdot \mathrm{cm}(I)^2 \leq (\Delta + \mathrm{cm}(I))^3$, this running time is of the form $f(\Delta + \mathrm{cm}(I)) \cdot |I|^{\mathcal{O}(1)}$ for a computable function f, so the problem is fixed-parameter tractable when parameterized by $\Delta + \mathrm{cm}(I)$. ◀

The two parameters of Theorem 8 enter the proof of Lemma 33 in different ways. The candidate edge multiplicity bounds every number $|\mathrm{Cand}(v)|$ and $|\mathrm{Cand}(u)|$ in the exponent of Lemma 32, whereas the maximum degree bounds only the number of children u of v, that is, the number of terms $|\mathrm{Cand}(u)|^2 + |\mathrm{Cand}(u)|$ of that exponent. By Theorems 4 and 5, respectively, neither parameter can be dropped. Section 6.4 nevertheless bounds the exponent of Lemma 32 without the maximum degree, by a single parameter, at least as large as the candidate edge multiplicity, that counts, for each vertex, the compatible pairs whose paths contain that vertex.

6.4 Bounded candidate vertex multiplicity

The maximum degree enters the running time of the algorithm of Section 6.3 only through the number of terms of the sum in the exponent of Lemma 32, one for each child of the vertex v of that lemma. In this subsection we bound that exponent without the maximum degree, by replacing the candidate edge multiplicity, which counts for each edge the compatible pairs whose unique paths use that edge, with the candidate vertex multiplicity of Section 4, which counts for each vertex the compatible pairs whose unique paths contain that vertex. Throughout this subsection, $I = (G, S, T, L)$ is a tree instance, Δ is the maximum degree of G, and the root r, the subtrees G_v, the boundary edges e_v, and the sets $\mathrm{Cand}(v)$ of candidate pairs are those of Section 6.3.

We first relate the candidate vertex multiplicity to the parameters of Theorem 8.

▶ **Observation 34.** *For every tree instance I on a tree of maximum degree Δ, $\mathrm{cm}(I) \le \mathrm{cvm}(I) \le \Delta \cdot \mathrm{cm}(I) + 1$.*

Proof. Every path that uses an edge contains both of its endpoints, so, for each edge $e = vw$ of G, every compatible pair counted in $\mathrm{cm}_I(e)$ is counted in $\mathrm{cvm}_I(v)$, and $\mathrm{cm}_I(e) \le \mathrm{cvm}_I(v) \le \mathrm{cvm}(I)$. This proves the first inequality. For the second, fix a vertex v and consider the compatible pairs whose paths contain v. At most one of them has a trivial path, namely the pair (v, v), since a trivial path containing v consists of the vertex v alone. The path of each of the remaining pairs uses at least one edge incident with v; assign to each of these pairs one such edge. A pair assigned an edge e has e on its path, so at most $\mathrm{cm}(I)$ pairs are assigned e, and at most Δ edges are incident with v. Hence, $\mathrm{cvm}_I(v) \le \Delta \cdot \mathrm{cm}(I) + 1$. ◀

By the second inequality of Observation 34, bounding the maximum degree together with the candidate edge multiplicity bounds the candidate vertex multiplicity, so Theorem 9 below implies the fixed-parameter tractability stated in Theorem 8. The algorithm is that of Section 6.3, unchanged, so by Lemma 31 the algorithm accepts if and only if I is a yes-instance. Only the bound on its running time changes. Lemma 32 bounds the time needed to compute the entries $\mathrm{DP}[v, X, \prec]$ of a vertex v in terms of $|\mathrm{Cand}(v)|$ and of $|\mathrm{Cand}(u)|$ for the children u of v, and the proof of Lemma 33 bounds these numbers by $\mathrm{cm}(I)$ and the number of children by Δ. Lemma 35 below bounds $|\mathrm{Cand}(v)|$ and the sum of the numbers $|\mathrm{Cand}(u)|$ by the candidate vertex multiplicity of v alone, however many children v has. Its proof uses only that the boundary edge e_v, when $v \neq r$, and the boundary edges e_u of the children u of v are all incident with v, so that the path of every pair of $\mathrm{Cand}(v)$ and of every pair of $\mathrm{Cand}(u)$ contains v.

▶ **Lemma 35.** *Let v be a vertex of G, and let u range over the children of v. Then $|\mathrm{Cand}(v)| \le \mathrm{cvm}_I(v)$ and $\sum_u |\mathrm{Cand}(u)| \le 2\,\mathrm{cvm}_I(v)$.*

Proof. If $v = r$, then $\mathrm{Cand}(v) = \emptyset$, and the first inequality holds. Otherwise, the boundary edge e_v is incident with v, so the path of every pair of $\mathrm{Cand}(v)$ uses e_v and therefore contains v. Hence, $\mathrm{Cand}(v)$ is a subset of the set of compatible pairs whose paths contain v, and this set has $\mathrm{cvm}_I(v)$ elements.

For each child u of v, the boundary edge e_u joins u to v, so the path of every pair of $\mathrm{Cand}(u)$ contains v as well. The path of a compatible pair is simple and therefore uses at most two edges incident with v, so every compatible pair belongs to $\mathrm{Cand}(u)$ for at most two children u of v. Hence, the sum $\sum_u |\mathrm{Cand}(u)|$ counts only compatible pairs whose paths contain v, of which there are $\mathrm{cvm}_I(v)$, and counts each of them at most twice, which proves the second inequality. ◀

▶ **Theorem 9.** *Single-Move Labeled Token Routing on a tree instance $I = (G, S, T, L)$ can be solved in time $2^{\mathcal{O}(\mathrm{cvm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$, where $\mathrm{cvm}(I)$ is the candidate vertex multiplicity of I. Consequently, the problem is fixed-parameter tractable when parameterized by $\mathrm{cvm}(I)$.*

Proof. We run the algorithm of Section 6.3, which accepts if and only if I is a yes-instance by Lemma 31, and we bound its running time. The paths of the compatible pairs and the sets $\mathrm{Cand}(v)$ are computed in time polynomial in $|I|$. Fix a vertex v of G, and let u range over the children of v. By Lemma 32, the entries $\mathrm{DP}[v, X, \prec]$ are computed in time $2^{|\mathrm{Cand}(v)|+\sum_u (|\mathrm{Cand}(u)|^2+|\mathrm{Cand}(u)|)} \cdot |I|^{\mathcal{O}(1)}$. By Lemma 35 applied to v, we have $|\mathrm{Cand}(v)| \leq \mathrm{cvm}_I(v) \leq \mathrm{cvm}(I)$ and $\sum_u |\mathrm{Cand}(u)| \leq 2\,\mathrm{cvm}_I(v) \leq 2\,\mathrm{cvm}(I)$, and by Lemma 35 applied to each child u of v, we have $|\mathrm{Cand}(u)| \leq \mathrm{cvm}_I(u) \leq \mathrm{cvm}(I)$. Therefore, $\sum_u |\mathrm{Cand}(u)|^2 \leq \big(\max_u |\mathrm{Cand}(u)|\big) \cdot \sum_u |\mathrm{Cand}(u)| \leq 2\,\mathrm{cvm}(I)^2$, and the exponent in the bound of Lemma 32 is at most $2\,\mathrm{cvm}(I)^2+3\,\mathrm{cvm}(I)$. Hence, these entries are computed in time $2^{\mathcal{O}(\mathrm{cvm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$, and summing over the n vertices of G yields the claimed running time. ◀

Although Theorem 9 implies the fixed-parameter tractability stated in Theorem 8, the two running times are incomparable. Substituting the bound $\mathrm{cvm}(I) \leq \Delta \cdot \mathrm{cm}(I) + 1$ of Observation 34 into the running time of Theorem 9 yields $2^{\mathcal{O}(\Delta^2 \cdot \mathrm{cm}(I)^2)} \cdot |I|^{\mathcal{O}(1)}$, which is worse than the running time of Theorem 8, whereas on trees of unbounded degree and bounded candidate vertex multiplicity only Theorem 9 applies. By the first inequality of Observation 34, bounding the candidate vertex multiplicity bounds the candidate edge multiplicity. By Theorem 9, no NP-hardness reduction to instances on trees can produce only instances of constant candidate vertex multiplicity, unless P = NP. The instances of Theorem 5 indeed have candidate edge multiplicity at most eight but not constant candidate vertex multiplicity: the literal pair and the witness pair of every occurrence have paths through the vertex g of that construction, so the candidate vertex multiplicity of these instances is at least twice the number of occurrences.

7 Conclusion and open problems

We introduced Single-Move Labeled Token Routing, in which every token moves at most once from its source to a compatible target. On unique-matching instances the problem is the discrete form of Monotone Sliding Reconfiguration [2], and we proved that it is NP-complete on planar graphs and on grids even for instances that, when solvable, admit a solution with pairwise edge-disjoint routes. Outside the unique-matching case, three hardness results hold on trees: NP-completeness with two colors on trees of maximum degree three, proved by Geft [3]; W[1]-hardness on trees of maximum degree three parameterized by an input bound on the solution edge multiplicity, which holds even when a movement order is given as part of the input; and NP-completeness on trees of unbounded degree whose candidate edge multiplicity is at most eight. The last two are proved in this paper, together with a linear-time algorithm on paths, a fixed-parameter algorithm parameterized by the number of tokens, and a fixed-parameter algorithm on trees, parameterized either by the maximum degree together with the candidate edge multiplicity or by the candidate vertex multiplicity alone; by the NP-completeness on trees of maximum degree three and on trees of candidate edge multiplicity at most eight, neither the maximum degree nor the candidate edge multiplicity can be omitted.

The two extreme compatibility relations remain the subject of open problems in robotics. Abrahamsen and Halperin list among ten open problems in geometric robotics the monotone reconfiguration of unit discs in the plane in which any disc may fill any target and a disc may

initially overlap a target, a geometric counterpart of our all-matching case [49]. At the other extreme, Geft, Halperin, and Nakar prove that unique-matching instances on $3 \times w$ grids remain NP-hard even when all sources lie to the left of all targets [2]. As soon as there are more than three tokens, their instances admit no solution with pairwise edge-disjoint routes, since every route uses one of the three edges that cross the line separating the sources from the targets. Theorem 3 gives hard instances on grids that admit such a solution, but its proof places a token whose source is its target at every vertex missing from the grid graph. Whether grids remain hard for instances that admit a solution with pairwise edge-disjoint routes and in which no vertex is both a source and a target is open. A second feature of the construction of Section 5.2, which Geft and Halperin identify as a possible source of hardness in their own constructions [50], is that a route may pass through unboundedly many sources and targets of other tokens: the route of an assignment token contains the two vertices created for every positive occurrence of its variable, or for every negative occurrence. Whether Theorem 3 holds when every route contains at most a constant number of sources and targets is open as well, whereas the construction of Theorem 5 has this property, since every path of a compatible pair contains at most two sources or targets other than its endpoints.

Further questions concern which measures of the structure of the graph and of the compatible pairs, alone or together, yield tractability. On trees, it remains to determine which restrictions on the compatible pairs yield tractability when one seeks a solution with small solution edge multiplicity. Beyond trees, bounding the pathwidth does not suffice, as the problem is NP-complete on $3 \times w$ grid graphs [2], whose pathwidth is at most three, and a graph of small treewidth may have arbitrarily many compatible pairs. On general graphs it would therefore be of interest to combine a measure of the structure of the graph with a measure of the compatibility relation, such as the number of compatible pairs separated by a small vertex set. Independently of such a combination, one may ask whether some structural parameter other than a measure of the graph's width yields fixed-parameter tractability by itself.

The compatibility relation itself raises two further directions. Call the compatibility relation a *coloring* when every source and every target carries exactly one label. The NP-completeness on trees of maximum degree three holds for two colors [3], whereas neither construction of this paper on trees is a coloring: a column tile token of Section 5.3 may move either to the tile subtree or to its local target, a choice token of Section 5.4 either to its selection target or to a non-selection target, and the second alternative is expressed by a second label. For Theorem 5 this label can be avoided: giving the three choice tokens of a clause one common color, merging their selection targets into one target of that color, giving the two non-selection targets the same color, and removing the filler tokens yields a coloring for which the reduction goes through unchanged. Whether Theorem 4 holds for colorings, and whether Theorems 4 and 5 hold for a bounded number of labels, is open. Conversely, is there a graph class or a parameter for which the problem is polynomial-time solvable or fixed-parameter tractable for colorings but not for arbitrary compatibility relations? A different restriction is to constrain the routes rather than the compatible pairs, by giving for each compatible pair a family of allowed routes, as in compatible-ordering problems [51].

Finally, two assumptions of the problem can be relaxed. If each token may move more than once, then a solution must also specify intermediate positions. On grid graphs, the unique-matching case remains NP-hard when each token may move at most twice [52]; we do not know whether Theorem 7 extends to this setting. If $|S| > |T|$, as in atom reconfiguration, then a solution consists of a set of compatible pairs in which every target occurs in exactly

one pair and every source in at most one pair, a route family, and a movement order; a token whose source is in no pair never moves, so the move of a pair is feasible only if its route contains no such source. Our hardness results carry over, since the instances with $|S| = |T|$ are exactly those in which every source is in a pair. Our algorithms carry over as well, since each of them constructs or enumerates the pairs of a solution and can leave a source out of every pair. The leaf-removal procedure of Theorem 1 leaves out a leaf that is a source but not a target whenever the current tree contains more sources than targets, the dynamic programming of Section 6.3 may leave out a source that is not a target when no path of a pair examined at that vertex contains it, without changing the running times of Theorems 8 and 9, and the algorithm of Theorem 7 also enumerates the sources left out. On a path, the order of the tokens never changes, so the sources left out can be chosen by dynamic programming along the path in polynomial time, although we do not know whether the linear running time of Theorem 6 can be preserved.

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